Core Finance
Week 8

MGMP 543

Prof. Benedict Guttman-Kenney

Spring 2025

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Welcome Back! Today's Topics

- Midterm
- Week 7 Recap
- Quiz 6
- New Content
  - MM Theory
  - Financing and Capital Structure
- Reminder: Boeing 7E7 Case Due Before Class Next Week. Bring Your Materials to Class. Be Prepared to Discuss the Case (Not Only The Assignment Questions)

Midterm

Midterm Exam Scores (120 was Maximum Score)

Section	Mean	Standard Deviation	25%	50% (Median)	75%
Combined	99	16	95	103	110
Monday (01)	96	18	87	104	110
Wednesday (02)	102	13	99	103	110

- See Canvas for your grade and solutions
- Some general advice
  - Read the question carefully and answer the question asked! Some answers were long / irrelevant.
  - Sense check your calculations. If I get $10,000 every other year forever, unlikely PV to be ~$5,000!

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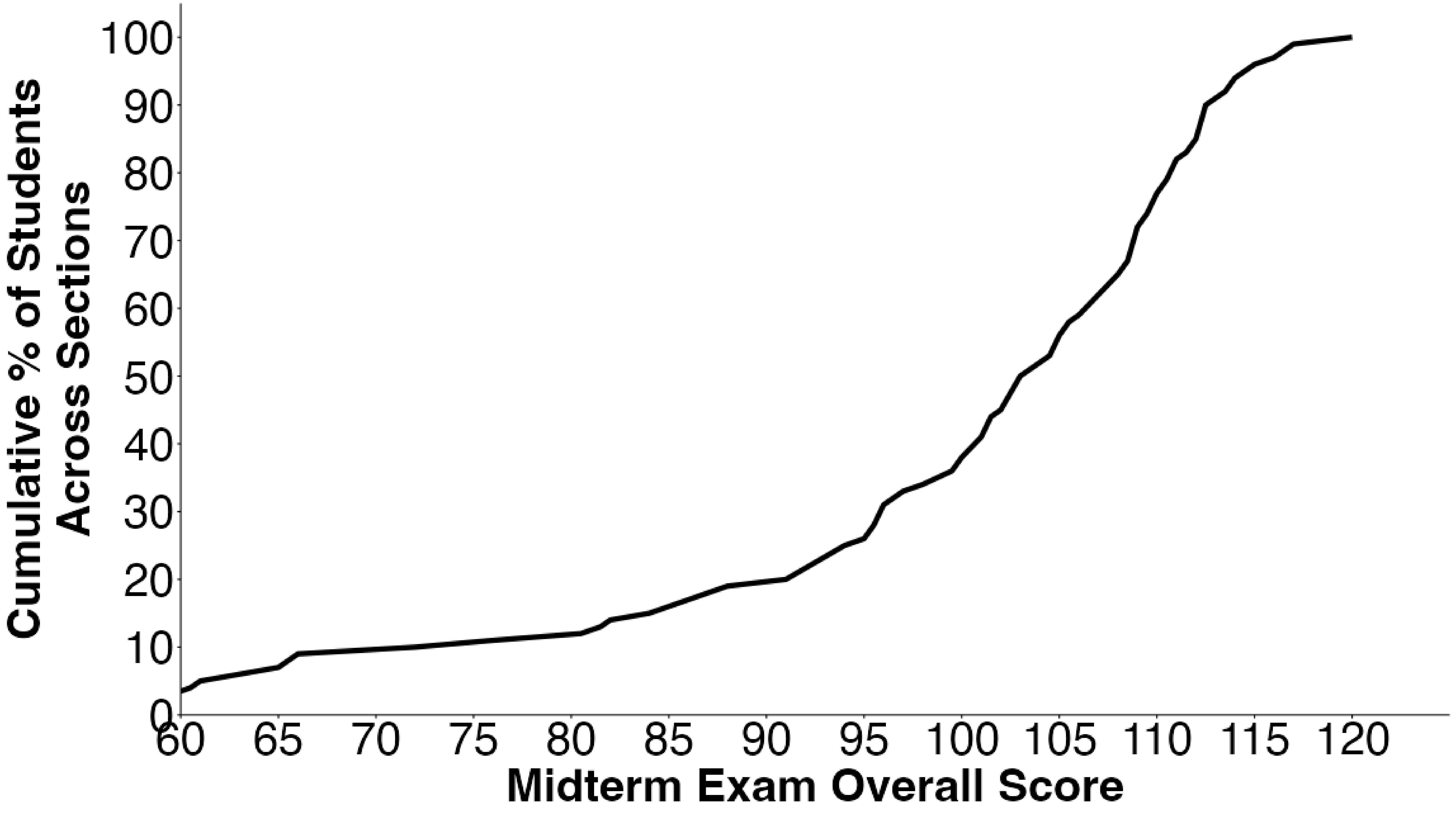
Midterm Exam Scores Across Both Sections



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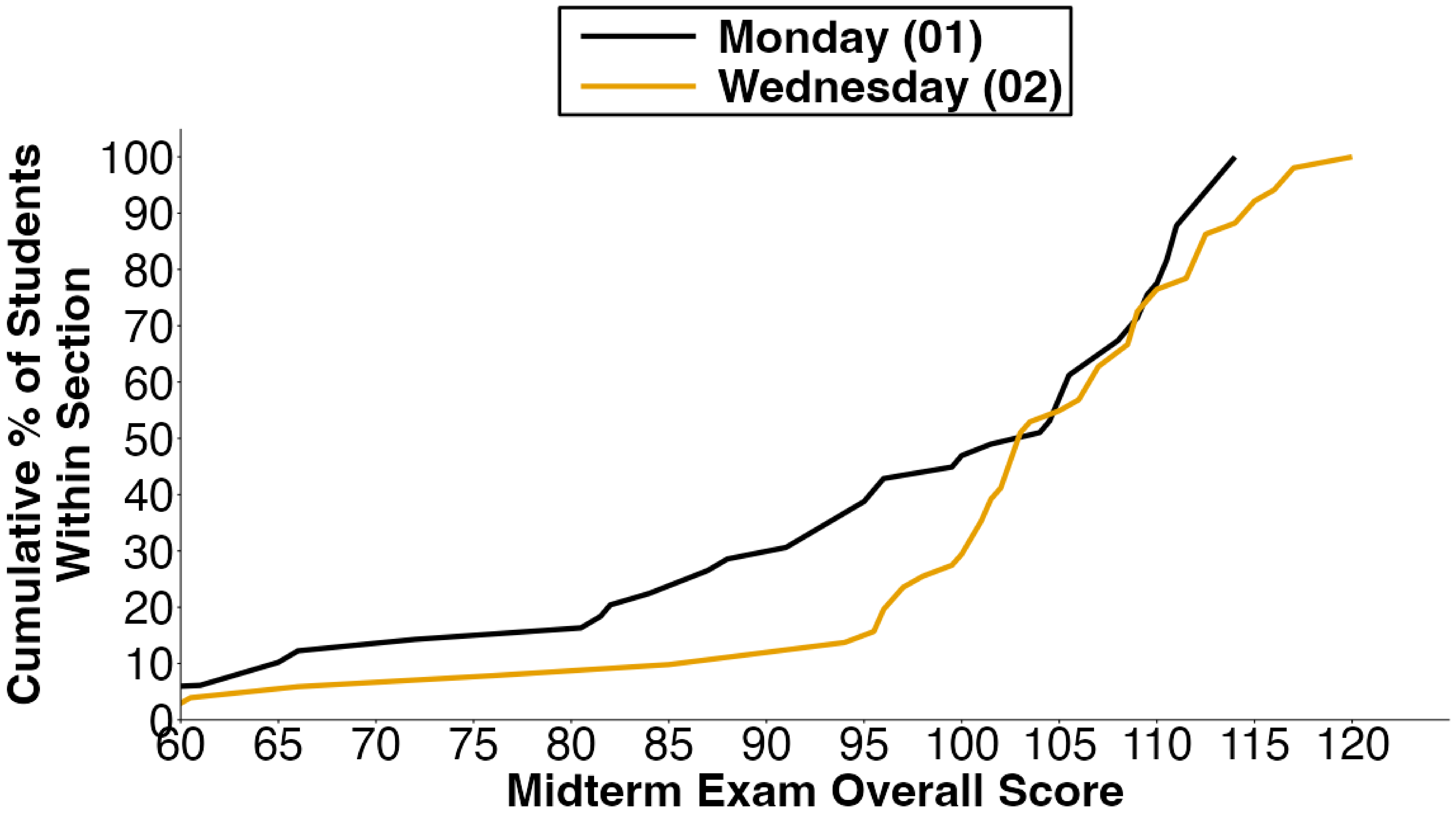
Midterm Exam Scores By Section



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Q13. Explain in 1-2 sentences what is incorrect in this statement "Higher inflation expectations have reduced the yield on long-term 30 year Treasury Bills."

- No such things as ‘long-term 30 year Treasury Bills’!
- Treasury Bills (T-Bills) are short-term, only for up to a year.
- Higher inflation expectations mean higher interest rate expectations which would increase the yields on Treasury Bills (or long-term Treasury Notes or Treasury Bonds).

Q16. "The price of a stock that pays dividends is the value of the assets in place plus the capitalized value of expected earnings under a non-growth policy." Is this statement true or false? Explain why in 1-2 sentences.

- False. The price of a stock is the value of assets in place PLUS capitalized value of earnings under a non-growth policy PLUS the growth options.
- The growth options weren't included.

Q18. You are taking out a loan. Would it be cheaper to borrow from (a) Laredo Loans at a 10% Annual Percentage Rate (APR) or (b) Frisco Financing at a 10% Effective Annual Rate (EAR)? Briefly explain in words or with a calculation. Assume interest is compounded monthly and all the other loan terms are the same between Laredo Loans and Frisco Financing.

- Cheaper to borrow at (b) from Frisco Financing 10% EAR.
- EAR for Laredo Loans at 10% APR compounded monthly would be
  $(1+ 0.1/12)^{12} – 1 = 10.47\%$
- This is more costly than (a) 10% EAR for Frisco financing.

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Q37. With a 10 percent annual interest rate, calculate the present value of receiving the B-PRIZE that awards $3,000 every other year forever, with the first payment 1 year from today.

- C/r is the normal perpetuity formula.
- But now we have a different timing of cash flows so you need to adjust the discount rate to go from a 1 year discount rate (given as 10%) to a 2 year discount rate:

(1+0.1)^2=0.21. PV = 3,000/0.21 = $14,285.71.

- But… this would be if the payments were in years 2, 4, 6, 8 etc. Instead, the payments are years 1, 3, 5, 7, 9 etc. So, we adjust for this by multiplying by the one-year difference in discount rates between years 1 and 2.

(3,000/0.21) x (1.1^2/1.1) = $15,714.29

- You could also have got to this in excel by putting the payments in excel and summing up the discounted cashflows.

Week 7 Recap

Week 7: What Did We Learn?

Week 7: What Did We Learn?

- 2 Nobel Prizes and Successful Financial Firms' Investment Strategies Based On Week 7 Content

Efficient Markets

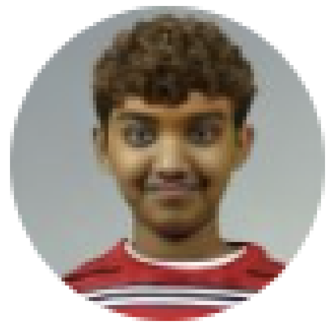
- Efficient Markets Hypothesis (what it is, how we get to efficient prices, implications)
- Weak-Form, Semi-Strong Form, Strong Form Efficiency
- Tests of Efficient Markets
- Active vs. Passive Investments. How Fees Affect Investments

Behavioral Finance (No free lunch but the price may not be right)

- Limits to Arbitrage (Fundamental Risk, Noise Trader Risk, Costs)
- Anomalies e.g., January Effect, Post-Earnings Drift, Shell, MCI, Palm, CUBA
- Puzzles (Equity Premium and Excess Volatility) and Momentum
- Retail Investor Have Biases, Overtrading Losses That Institutional Investors Benefit From
- "Sophisticated" CEOs and Traders Also Have Behavioral Biases.
- Firms Made Up of Humans So Psychology Matters!

What Are We Learning In Real-Time?

The Value of Compounding



Young Parik
@YoungParikPatel

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The nice thing about three consecutive 4% drops is that each 4% is lower in real value than the one that proceeded it. That’s the power of compounding

7:24 PM · Apr 6, 2025 · **309.3K** Views

What Are We Learning In Real-Time?

Information is Quickly Incorporated into Prices…This Can Include Incorrect Information

What Are We Learning In Real-Time?

The Value (?) of Bitcoin, Private Equity, and Hedge Funds 🧐

Mike Bird @Birdyword · 22h
Almost timed to perfection

Tyler Winklevoss @tyler · Apr 6
For the first time in history, bitcoin is not moving in lockstep with the stock market. It's now behaving like a hedge to geopolitical uncertainty. When the stock market plunged during Covid, so did bitcoin. And this was always case over the last 10+ yrs. But not this time.

Readers added context

To no fault of OP's own, Bitcoin began sharply declining moments after this was posted. By around 9:00PM ET, it had declined by around 7%, from around $82,300 to $78,000 USD.

coinmarketcap.com/currencies/bit...

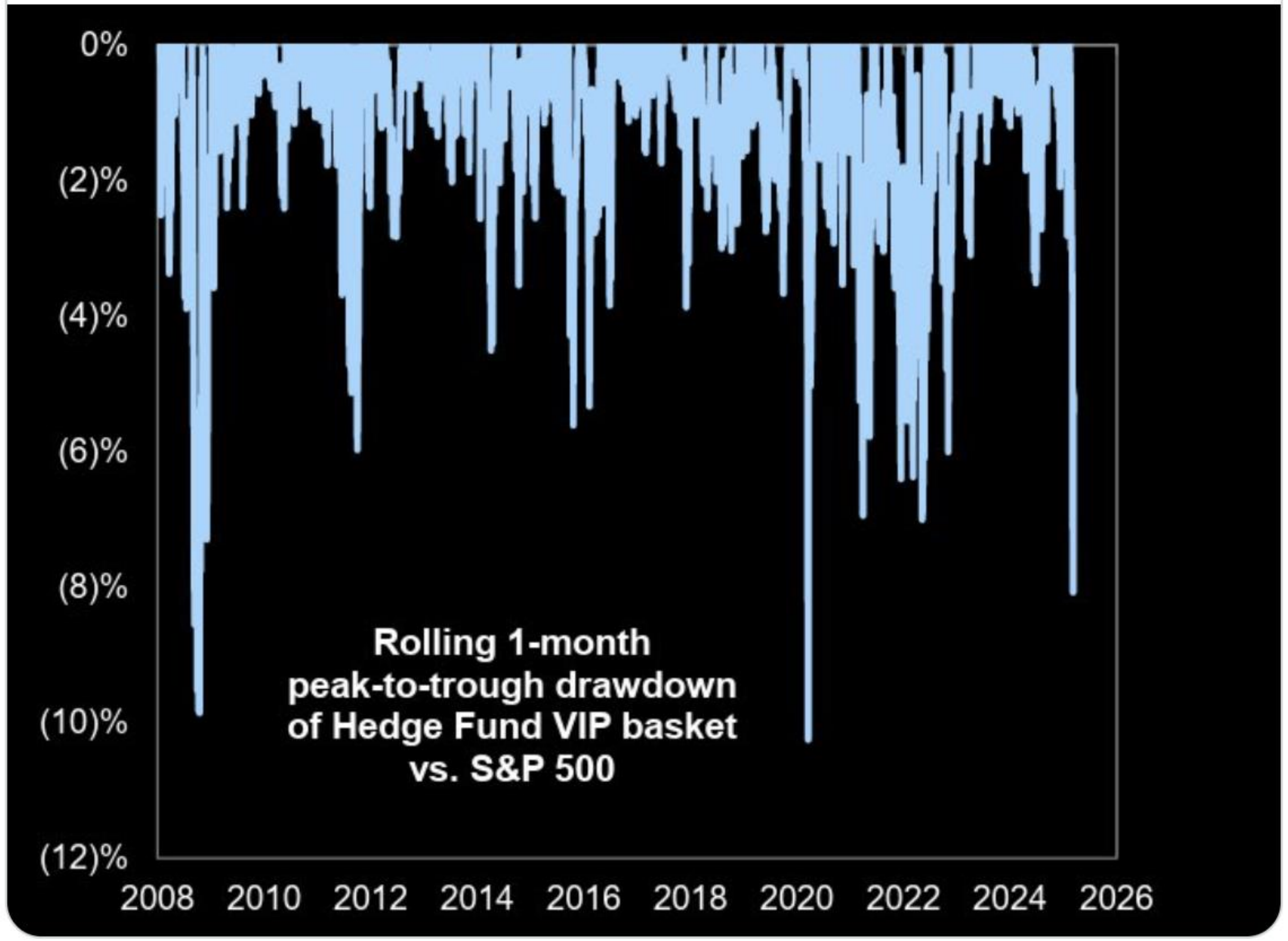
Do you find this helpful? Rate it

You reposted
Arpit Gupta @arpitrage
Really rough for private funds, with all the big PE names down 20-40% this year

Blackstone Inc	$119.67	↓30.59%
KKR & Co Inc	$90.53	↓38.80%
Apollo Global Manage...	$107.27	↓35.05%
Carlyle Group Inc	$35.56	↓29.57%
Brookfield Corp	$45.05	↓21.58%

Arpit Gupta @arpitrage
hedge funds don't give you as much upside in good times, they make up for it by crashing in bad times

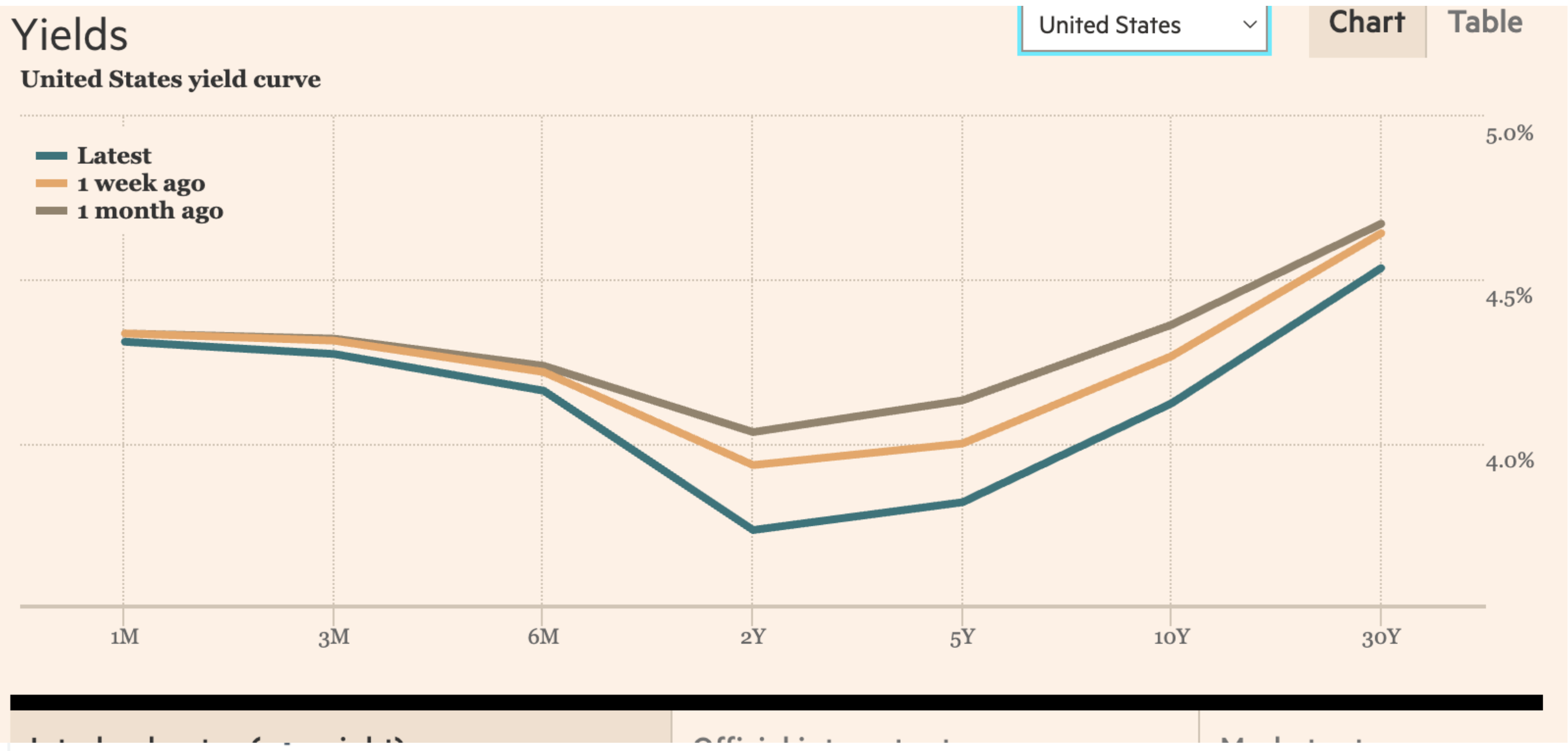
Yaron Naymark @1MainCapital · Apr 2
One of the worst periods for HFs in the past 15 years



7:34 AM · Apr 2, 2025 · **3,326** Views

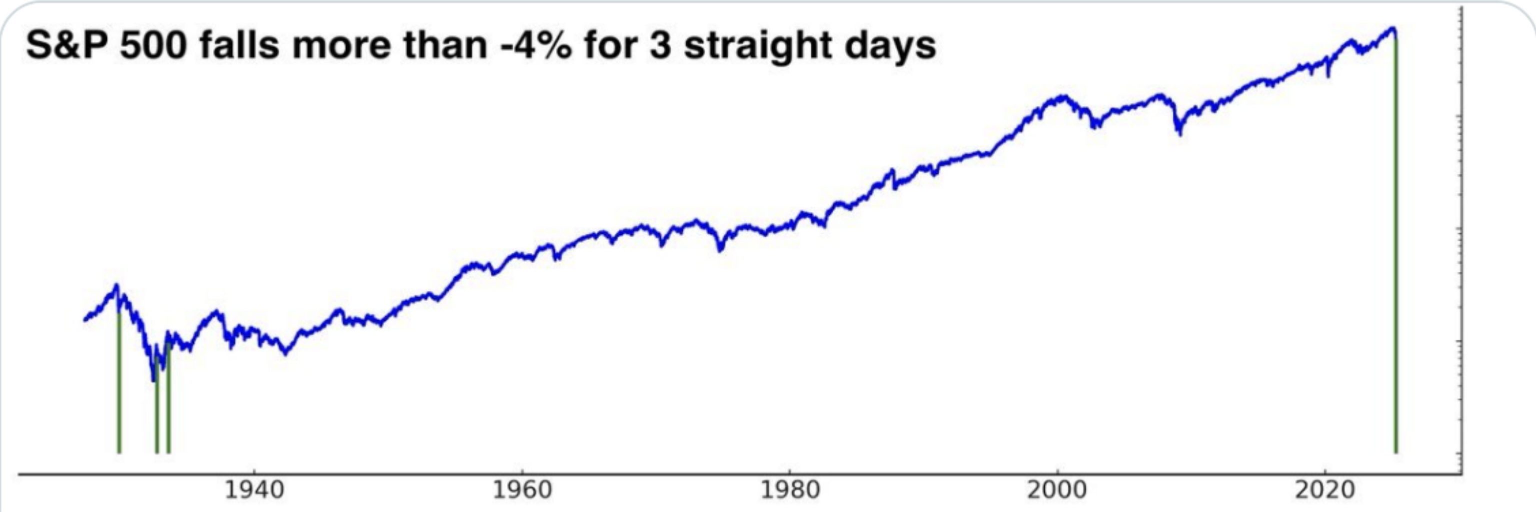
What Are We Learning In Real-Time?

Markets View Tariffs / Policy Uncertainty Negatively



Maine @TheMaineWonk

If tomorrow, the S&P 500 falls more than 4% for the third day in a row, it'll be for the first time since the great depression.



S&P 500 after 3 straight days of -4% declines
@SubuTrade

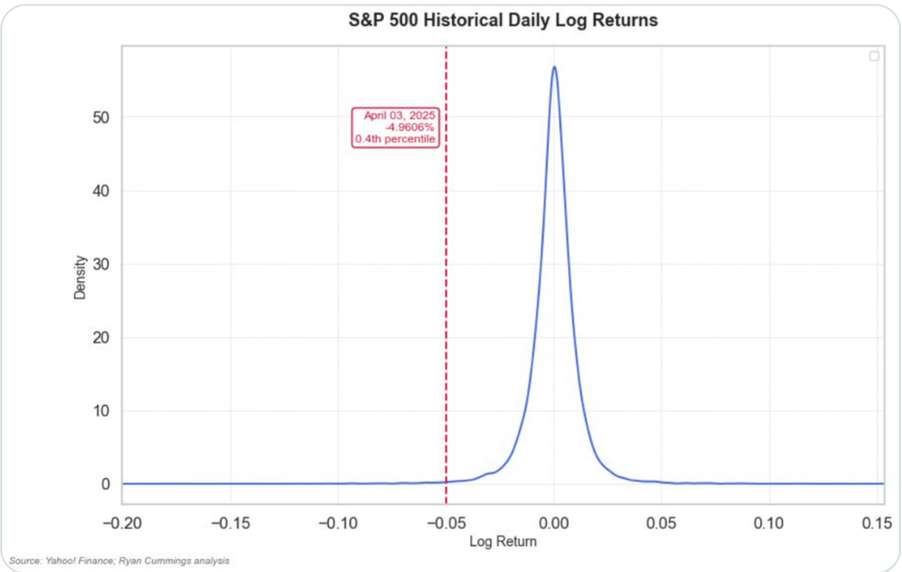
	1 Day Later	2 Days Later	3 Days Later	4 Days Later	1 Week Later	2 Weeks Later	3 Weeks Later
November 13, 1929	8.95%	14.95%	13.76%	17.21%	20.44%	18.46%	27.12%
September 14, 1932	4.22%	2.99%	2.04%	-0.14%	3.67%	9.39%	10.88%
July 21, 1933	0.62%	8.81%	6.94%	9.74%	10.67%	7.36%	10.16%
April 7, 2025?							

6:45 PM · Apr 6, 2025 · **1M** Views

Ryan Cummings @weakinstrument

If you do this since 1929, it's even worse.

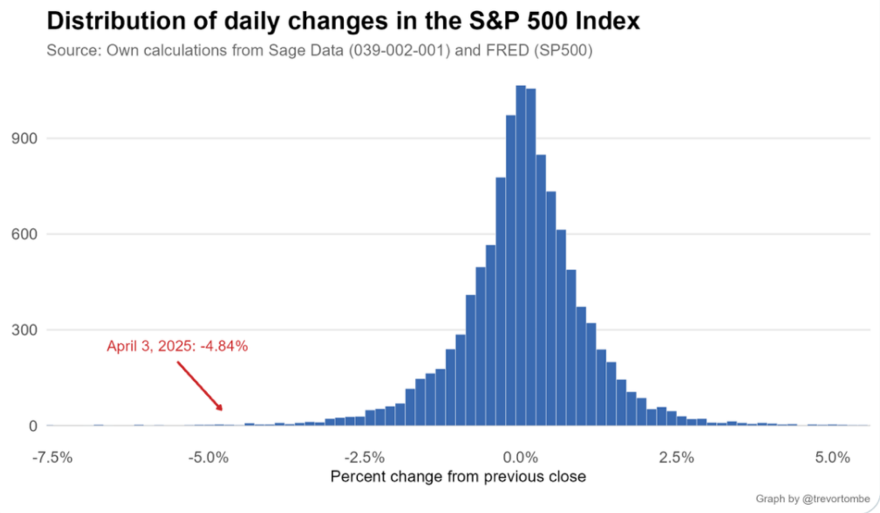
Today was worse than 99.6% of trading days since 1929. It was the 95th worst day of all time (out of >24,000 days), and the 35th worst day since 1945.



Trevor Tombe @trevortombe · Apr 3

How bad was today's stock market decline?

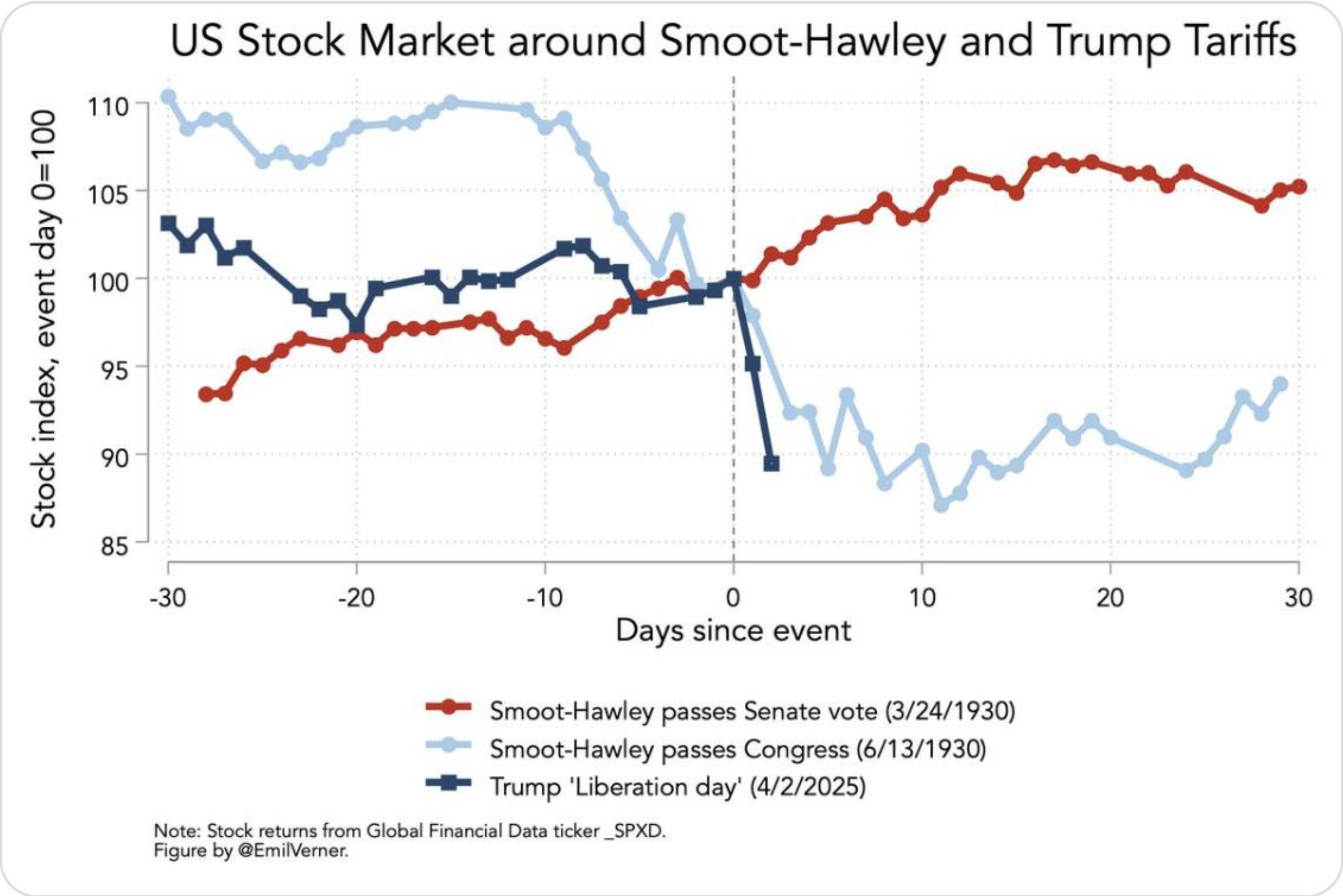
Of the 11,406 trading days since January 1980, only 29 had larger declines.



5:58 PM · Apr 3, 2025 · **245.6K** Views

Emil Verner @EmilVerner

US stock market around Smoot Hawley passage and last week's Trump tariffs



8:31 PM · Apr 6, 2025 · **13.2K** Views

Exclusive: Klarna Pauses Planned IPO in Wake of Trump Tariff Announcement

Quiz 6

Q2. In order to test the strong form of market efficiency, researchers have examined the

a) Recommendations of professional security analysts.
b) Recommendations of professional security analysts, performance of mutual funds, and performance of pension funds.
c) Performance of mutual funds and performance of pension funds.
d) Recommendations of professional security analysts and performance of mutual funds.

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c) Performance of mutual funds and performance of pension funds.
d) Recommendations of professional security analysts and performance of mutual funds.

Q3. Which of the following observations would provide evidence against the strong form of efficient market theory?

a) Mutual fund managers do not on average make superior returns.
b) Managers who trade in their own firm's stocks make superior returns.
c) In any year, approximately 50 percent of all pension funds outperform the market.
d) Mutual fund managers do not on average make superior returns and, In any year, approximately 50 percent of all pension funds outperform the market.

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Q4. Suppose that a lawyer works for a firm that advises corporate firms planning to sue other corporations for antitrust damages. He finds that he can "beat the market" by short selling the stock of firms that will be sued. This hypothetical finding would violate the

a) Semi-strong form hypothesis of market efficiency
b) Weak-form hypothesis of market efficiency
c) None of the hypotheses of market efficiency
d) Strong-form hypothesis of market efficiency

Q4. Suppose that a lawyer works for a firm that advises corporate firms planning to sue other corporations for antitrust damages. He finds that he can "beat the market" by short selling the stock of firms that will be sued. This hypothetical finding would violate the

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c) None of the hypotheses of market efficiency
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Q5. If the efficient market hypothesis holds, investors should expect

a) To receive a fair price for their security
b) To be able to pick stocks that will outperform the market
c) To receive a fair price for their security and to earn a normal rate of return on their investments
d) To earn a normal rate of return on their investments

Q5. If the efficient market hypothesis holds, investors should expect

a) To receive a fair price for their security
b) To be able to pick stocks that will outperform the market
c) To receive a fair price for their security and to earn a normal rate of return on their investments
d) To earn a normal rate of return on their investments

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Capital Structure of a Firm


Great! Where are we going?

Aim: Value any project/firm

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- Weeks 1-5: Tools for valuing assets
- Week 6: Hopefully, prices are right 😅
- Week 8: Compare financing options
- Week 9: Putting this together (with some accounting) to value any project/firm!

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Why Do Firms Issue Securities (Debt/Equity)?

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- Answer: To fund NPV > 0 projects! (Hopefully not to fund NPV <0 projects!)

- Today, we consider whether to finance projects using:
    - Debt
    - Equity
    - Mixture of debt & equity

Why Do Firms Issue Securities (Debt/Equity)?

- Answer: To fund NPV > 0 projects! (Hopefully not to fund NPV <0 projects!)
- Today we investigate whether to finance projects using:
  - Debt
  - Equity
  - Mixture of debt & equity
- Which ‘capital structure’ (% of debt vs. equity) would ⬆ a firm’s value?

Benefits & Costs of Financing With **Equity**

Benefits

- ⬆️ Financial flexibility in the future
- Aligned incentives

Costs

- ‘Expensive’ – offering part of the upside of a project
- Loss of Control – shareholders have voting rights

Benefits & Costs of Financing With **Debt**

Benefits

- Interest on debt is tax deductible (whereas dividends is not tax deductible)
- Discipline on management (limit empire building), enables direct monitoring & control

Costs

- ⬆ probability of financial distress
  - direct and indirect bankruptcy costs, ⬇ financial flexibility
- Potential conflict between bondholders and shareholders

MM Theory: No Taxes

Another Nobel Prize Winning Lecture: (**M**odigliani and **M**iller) **MM Theory** (1958)

The Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel 1985

Photo from the Nobel Foundation archive.

Franco Modigliani

Prize share: 1/1

The Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel 1990

Photo from the Nobel Foundation archive.

Harry M. Markowitz

Prize share: 1/3

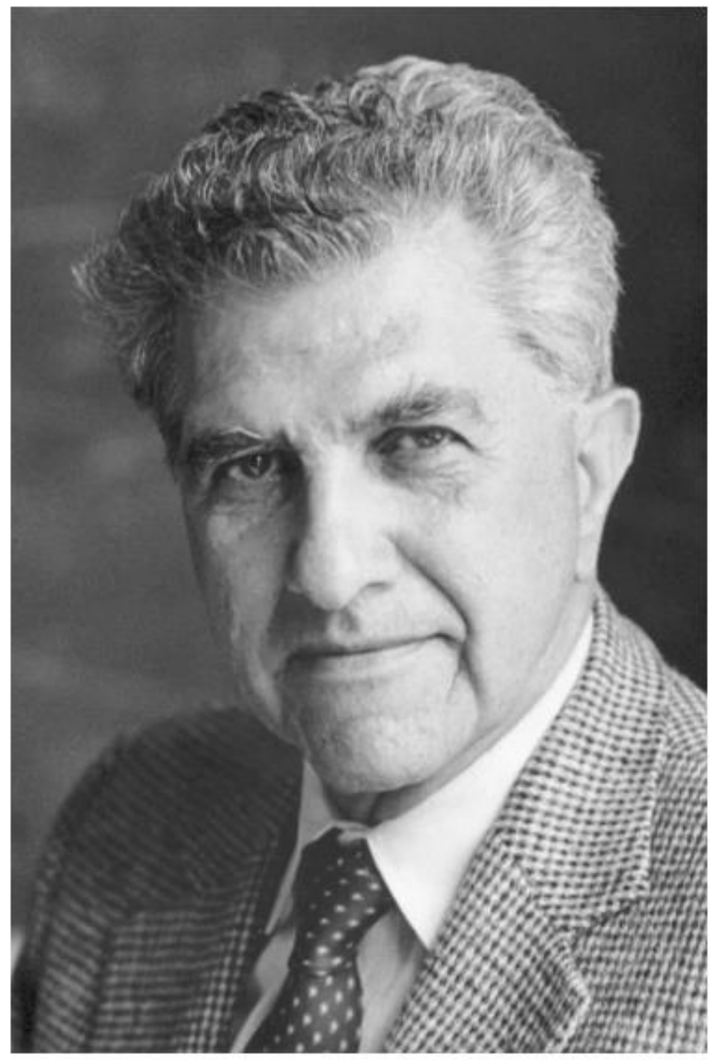


Photo from the Nobel Foundation archive.

Merton H. Miller

Prize share: 1/3

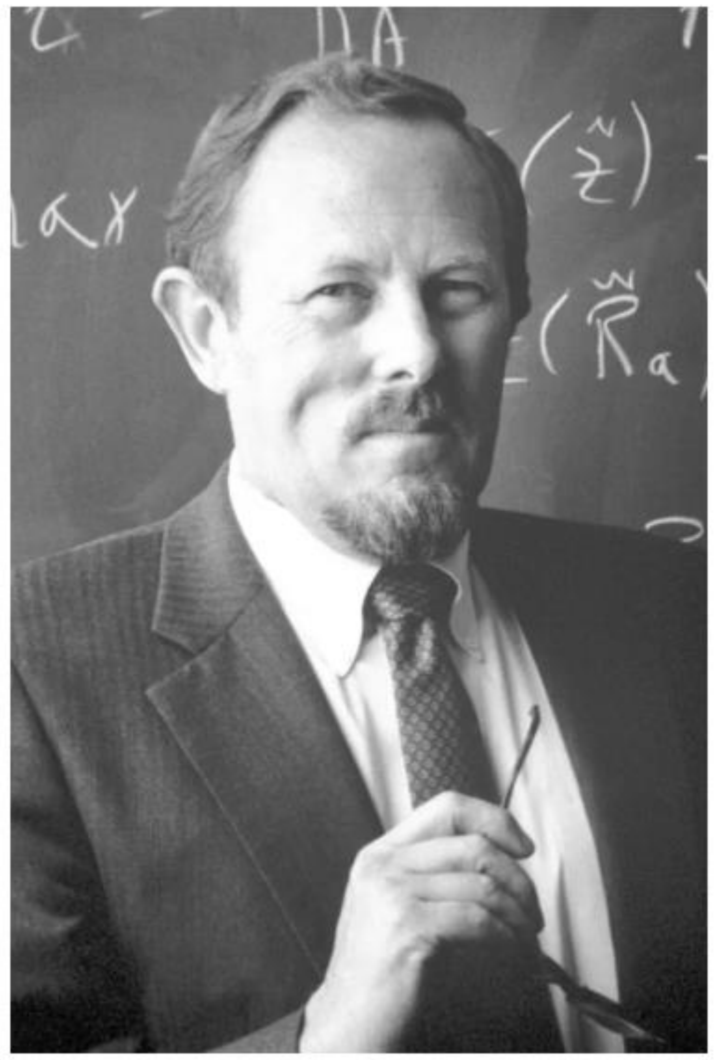


Photo from the Nobel Foundation archive.

William F. Sharpe

Prize share: 1/3

MM Theory

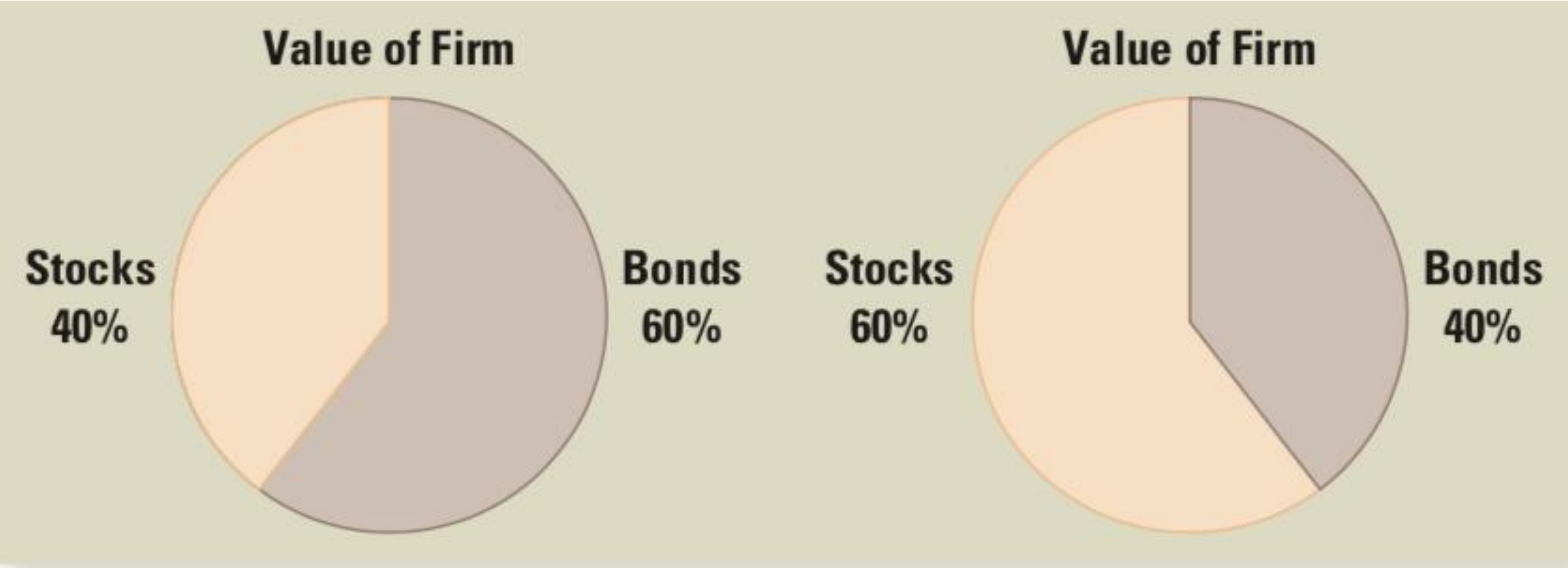
- MM Proposition 1: **Capital Structure** is **irrelevant** to the value of a firm (also known as the law of conservation of value)
- MM Proposition 2: Required **Rate of Return** to equity holders rises with leverage

- These hold under some assumptions…

MM Theory: No Taxes

MM Proposition 1: **Capital Structure** is **irrelevant** to the value of a firm

- The firm's value is determined by the value of its real assets (left-hand-side of balance sheet) NOT by the % of debt or equity issued to buy the assets.
- The overall pie (value of firm) is the same size!



MM Proposition 1: **Capital Structure** is **irrelevant** to the value of a firm

- The firm's value is determined by the value of its real assets (left-hand-side of balance sheet) NOT by the % of debt or equity issued to buy the assets.
- The overall pie (value of firm) is the same size!
- Value of an asset is preserved irrespective of how the claims are made against it.
- Total value of a levered firm (debt + equity), V_L,
  is equal to the value of an all equity (unlevered) firm, V_U

$$V_U = V_L = D + E$$

$$PV(D) + PV(E) = PV(D + E) = PV(A)$$

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MM Proposition 2: Required **Rate of Return** to equity holders rises with leverage

- Debt-to-equity ratio (D/E) is a measure of **leverage**

  calculated using **market values** (*not* book values)
- Total firm risk is the same irrespective of capital structure

  i.e., firm's *overall* cost of capital remains unchanged.
- If increase debt, and so reduce equity, then total risk is shared among fewer equity holders. So, the risk on equity holders must increase, and so demand a higher return.

Weighted-Average Cost of Capital (WACC) - More On This Later Today!

WACC is the opportunity cost of capital for firm, and is constant across D/E ratios:

$$r_A = \left(\frac{D}{D+E} \times r_D\right) + \left(\frac{E}{D+E} \times r_E\right)$$

Rearrange for the equity return:

$$r_E = r_A + (r_A - r_D)\frac{D}{E}$$

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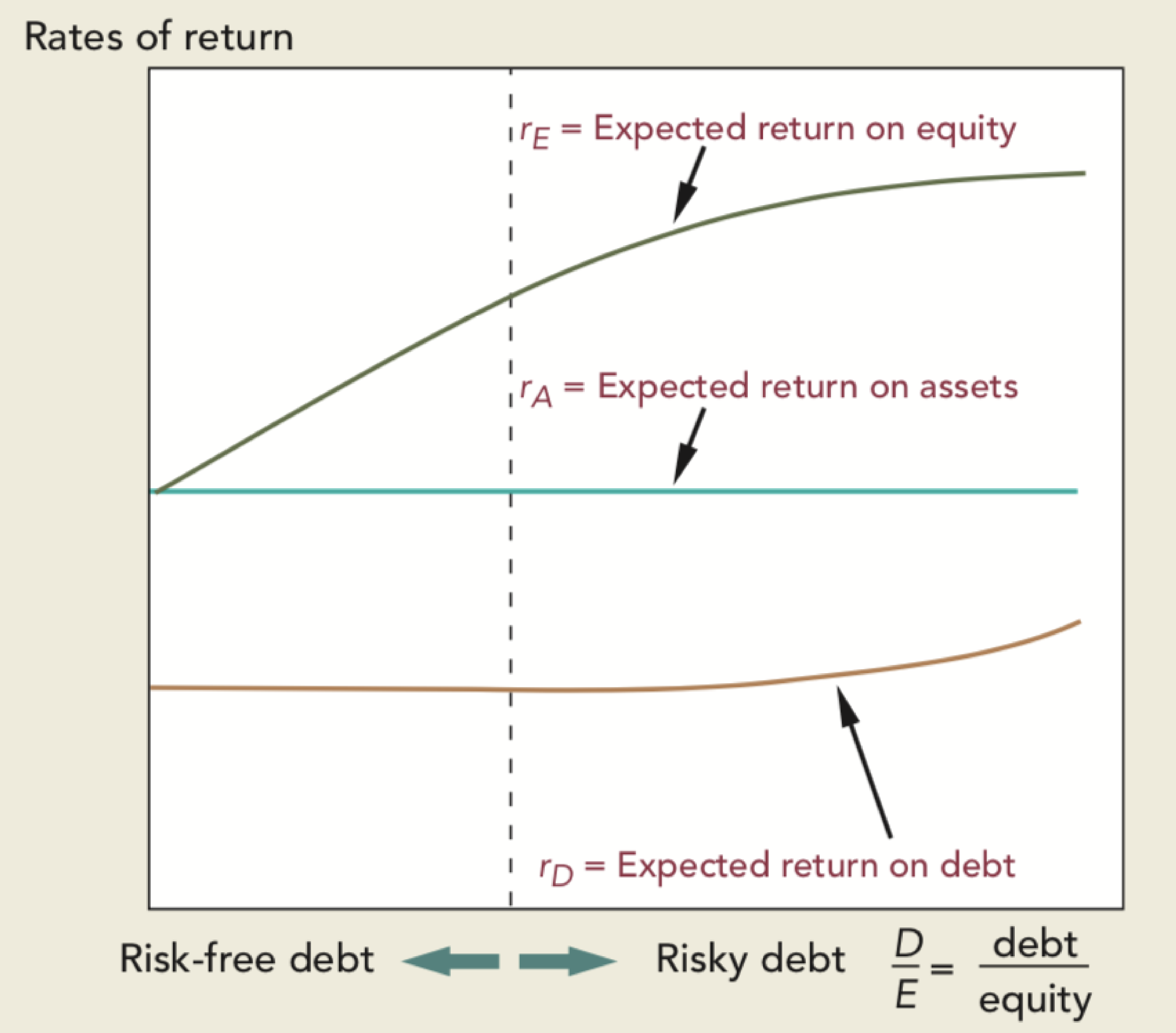
MM Proposition 2: Required **Rate of Return** to equity holders rises with leverage. The rate of the increase depends on the spread between r_A, the required rate of return on a portfolio of all the firm's securities, and r_E, the required rate of return on debt.

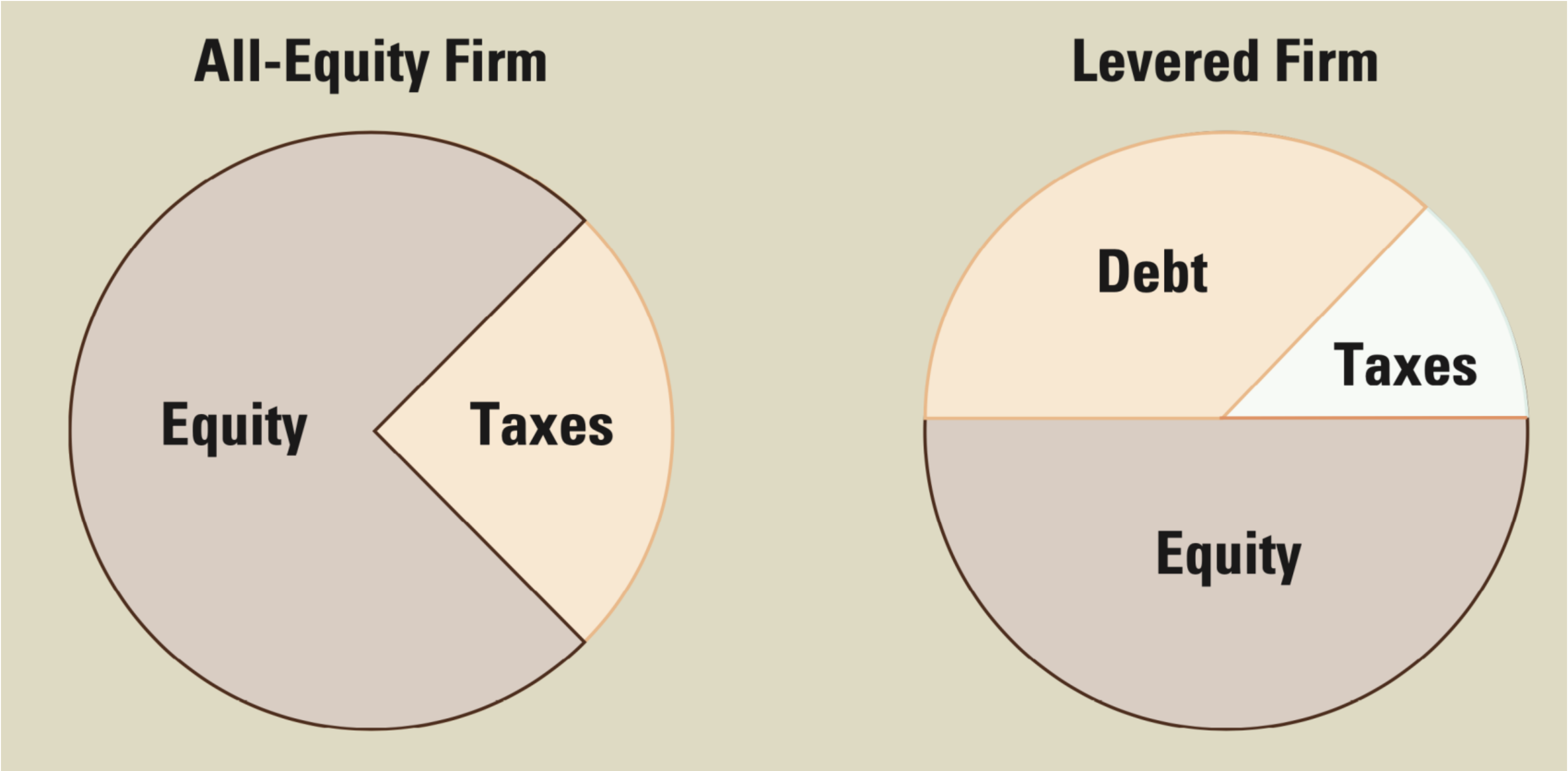


$$r_E = r_A + (r_A - r_D)\frac{D}{E}$$

$$r_A = \left(\frac{D}{D+E} \times r_D\right) + \left(\frac{E}{D+E} \times r_E\right)$$

MM Theory: With Taxes

MM Proposition 1 (With Taxes). The upside of debt financing



Without taxes:

- Total value of a levered firm (debt + equity), V_L, is equal to the value of an all equity (unlevered) firm,

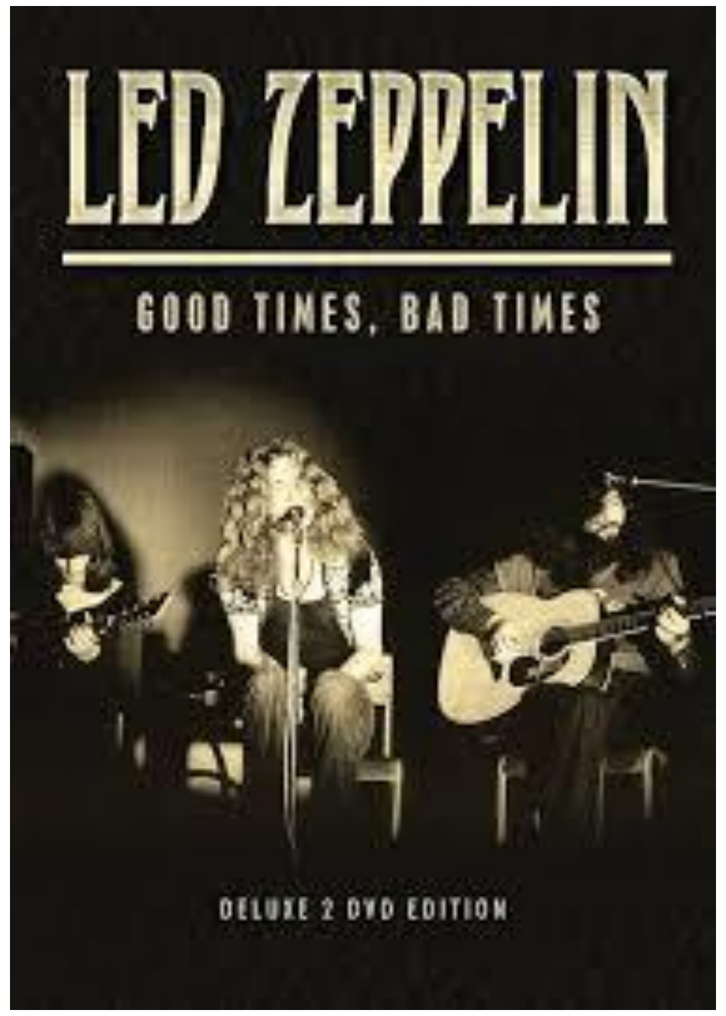
$$V_U = V_L = D + E$$

With taxes:

- Now, $V_L = V_U + T_C D$, where T_C is the marginal corporate tax rate.
- Present Value of the '**Debt Tax Shield**' $T_C D = V_L - V_U$

  i.e., the difference between levered firm and unlevered firm
- Now there is a **corporate tax advantage to issuing debt** (instead of equity), as the interest payments on debt reduce the pre-tax income of the firm 🙂

MM Proposition 1 (With Taxes)
The downside of debt financing

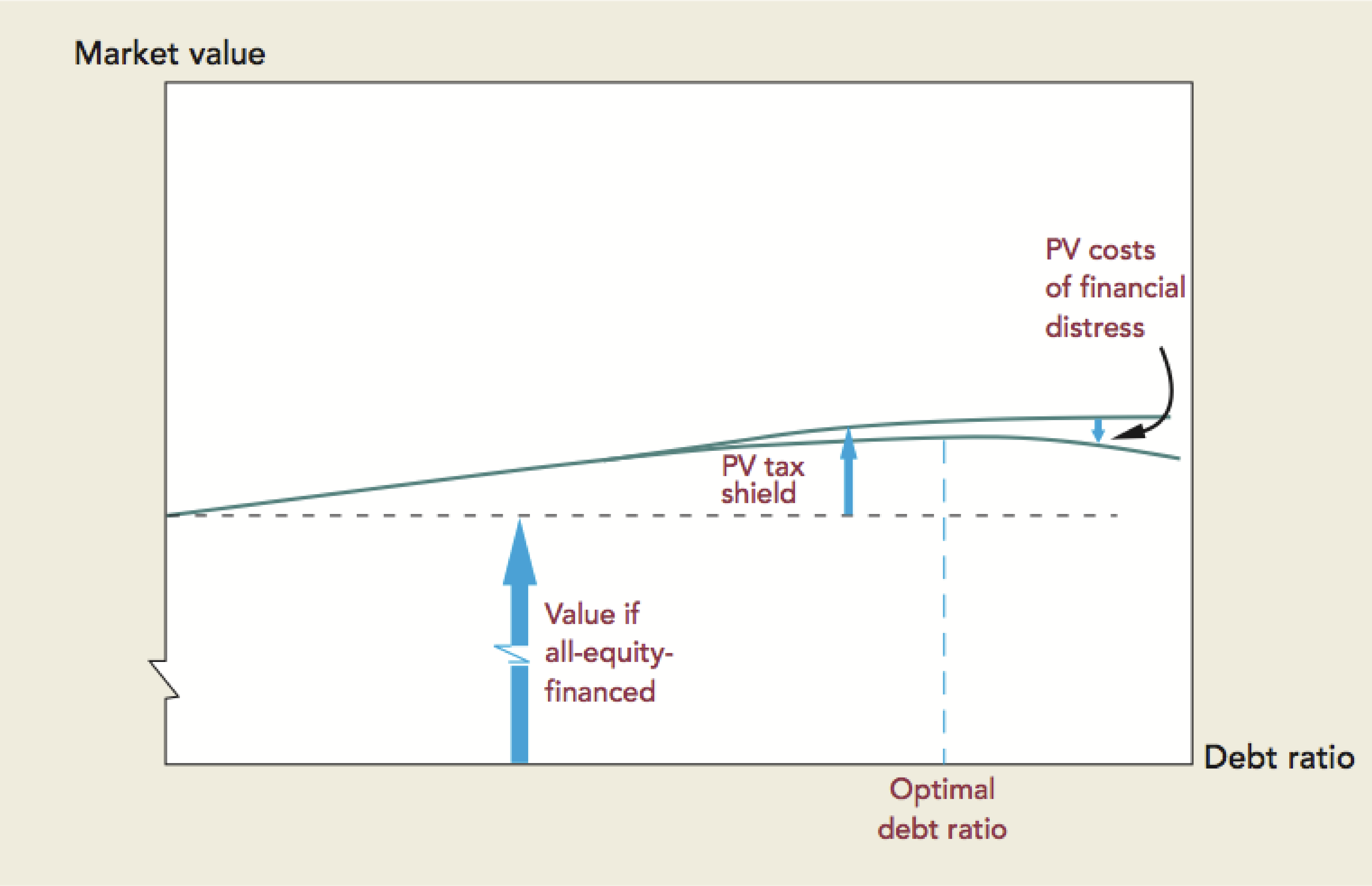


- Levered firms are at increased risk of (costly) financial distress 🙁
  - Why?
  - A levered firm needs to deliver a fixed amount of cashflows to service debt interest payments *irrespective* of economic conditions / firm profitability.
  - So in bad times, a levered firm may struggle to do this and lead to bankruptcy/restructuring.
  - Whereas an unlevered firm could just reduce or not pay dividends in bad times.
  - '**Debt Overhang**' describes making debt payments is hindering future investments.
- Investors know this and so it limits a firm's ability to issue debt. 🙁
- PV of financial distress is hard to measure (more on this later today!).

Total Value = All Equity Value + PV(Tax Shield) - PV(Financial Distress)

MM Proposition 1 (With Taxes)
The trade-off of debt financing

Total Value = All Equity Value + PV(Tax Shield) - PV(Financial Distress)



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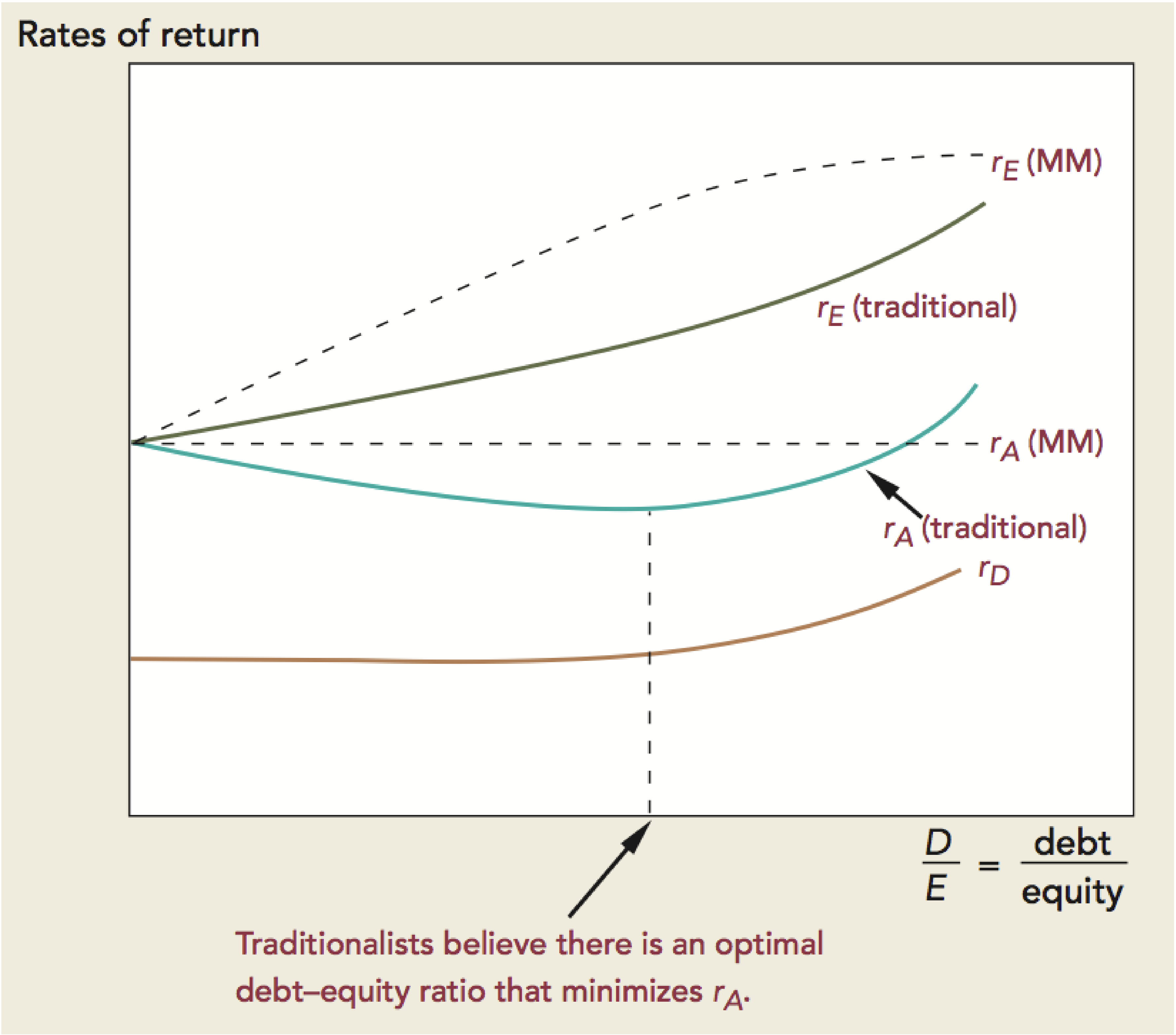
MM Proposition 2 (With Taxes): Required **Rate of Return**

- Now updating our after-tax WACC:

$$WACC = (1 - T_c)\, r_D \left(\frac{D}{V}\right) + r_E \left(\frac{E}{V}\right)$$

- If firm receives a tax shield on their interest payments, then the after-tax WACC declines as debt increases!
- Cost of Equity (r_E), where r_A if all-equity firm

$$r_E = r_A + \frac{D}{E}(r_A - r_D)(1 - T_c)$$



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MM Proposition 2 (With Taxes): Required **Rate of Return**

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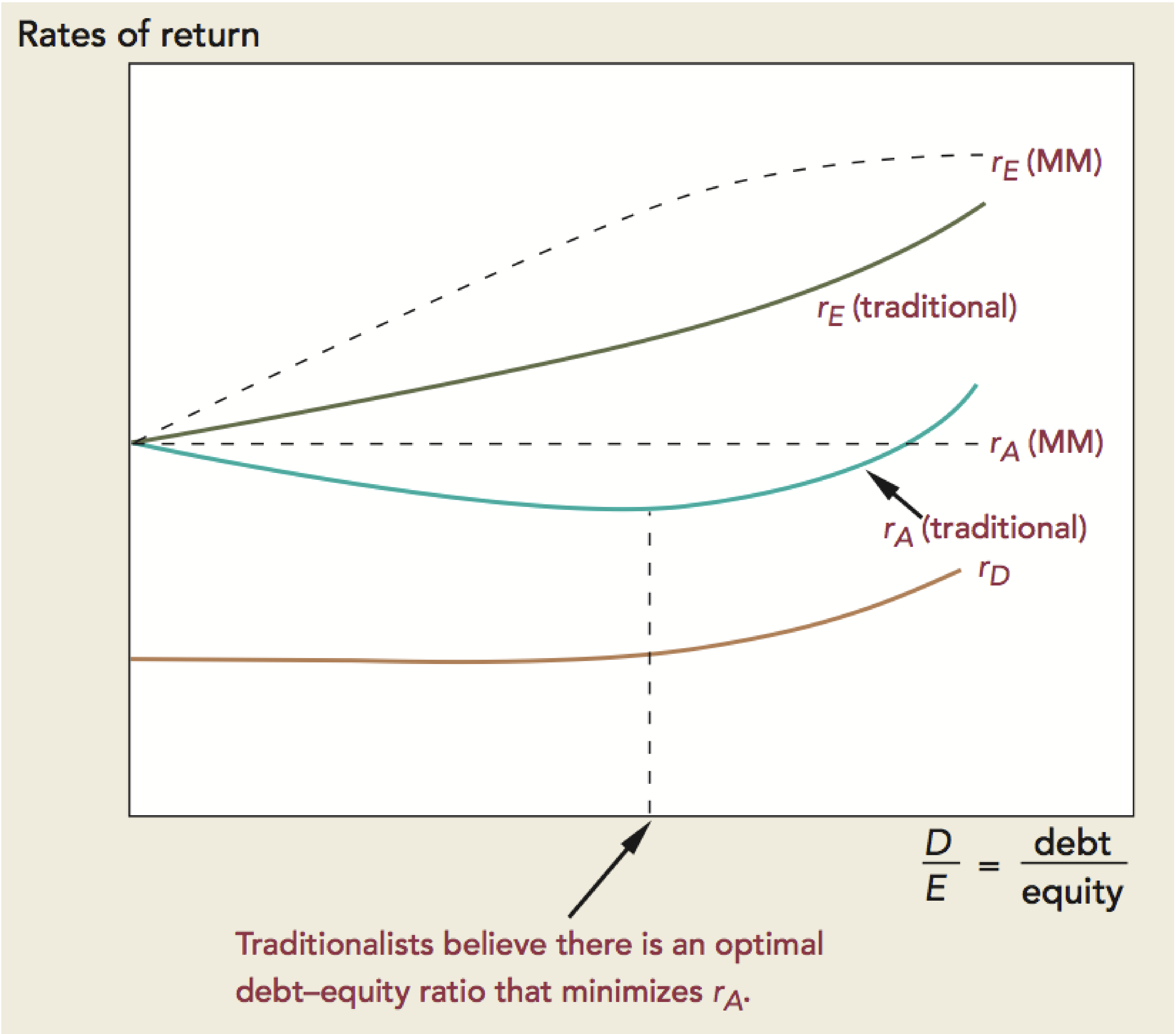
- If firm receives a tax shield on their interest payments, then the after-tax WACC declines as debt increases! Optimal is 100% debt.
- Cost of Equity for levered firm (r_E)

$$r_E = r_A + \frac{D}{E}(r_A - r_D)(1 - T_c)$$

n.b. r_A is cost of equity for an all-equity firm

- Can write in terms of betas:

$$\beta_E = \beta_A(1 + (1 - T_c)\frac{D}{E})$$



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MM Theory Assumptions

1. Capital markets are frictionless
2. No personal taxes
3. No corporate taxes (for MM Theory without taxes)
4. No costs to bankruptcy
5. Borrow and lend at the risk-free rate
6. Everyone has the same information
7. Managers maximize shareholders' wealth

These assumptions do not hold in the real-world. 🙁
But…MM Theory gives us a framework to think about capital structure 🙂
& why departures from these assumptions influence real-world decisions 😐

WACC

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Aim: Calculate Borrowing Cost Of The Entire Firm

Weighted Average Cost of Capital (WACC):

$$WACC = (1 - T_c)\, r_D \left(\frac{D}{V}\right) + r_E \left(\frac{E}{V}\right)$$

Aim: Calculate Borrowing Cost Of The Entire Firm

Weighted Average Cost of Capital (WACC):

$$WACC = (1 - T_c)\, r_D \left(\frac{D}{V}\right) + r_E \left(\frac{E}{V}\right)$$

What about if firm has **preferred shares**, (as well as common shares, and debt):

$$WACC = (1 - T_c)\, r_D \left(\frac{D}{V}\right) + \boldsymbol{r_P} \left(\frac{\boldsymbol{P}}{\boldsymbol{V}}\right) + r_E \left(\frac{E}{V}\right)$$

Aim: Calculate Borrowing Cost Of The Entire Firm

Weighted Average Cost of Capital (WACC):

$$WACC = (1 - T_c)\, r_D \left(\frac{D}{V}\right) + r_E \left(\frac{E}{V}\right)$$

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Should we use book value or market value in these calculations?

Aim: Calculate Borrowing Cost Of The Entire Firm

Weighted Average Cost of Capital (WACC):

$$WACC = (1 - T_c)\, r_D \left(\frac{D}{V}\right) + r_E \left(\frac{E}{V}\right)$$

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$$WACC = (1 - T_c)\, r_D \left(\frac{D}{V}\right) + \boldsymbol{r_P} \left(\frac{\boldsymbol{P}}{\boldsymbol{V}}\right) + r_E \left(\frac{E}{V}\right)$$

Should we use book value or market value in these calculations? **MARKET VALUE!**

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What About Valuing Projects?

- A project's cost of capital depends on ***how*** the capital is being used (the project's risk) ***not*** the source of the company (the company's risk).
  - **Why? Opportunity cost of capital – see week 5 SML, the risk/return of that capital elsewhere**
  - If a firm uses **one** discount rate for all projects, it may increase the company's risk & reduce value.

What About Valuing Projects?

- A project's cost of capital depends on ***how*** the capital is being used (the project's risk) ***not*** the source of the company (the company's risk).
  - **Why? Opportunity cost of capital – see week 5 SML, the risk/return of that capital elsewhere**
  - If a firm uses **one** discount rate for all projects, it may increase the company's risk & reduce value.
- Similarly, for conglomerates use the cost of capital for the relevant business, not overall cost of capital.

What About Valuing Projects?

- A project's cost of capital depends on ***how*** the capital is being used (the project's risk) ***not*** the source of the company (the company's risk).
  - **Why? Opportunity cost of capital – see week 5 SML, the risk/return of that capital elsewhere**
  - If a firm uses **one** discount rate for all projects, it may increase the company's risk & reduce value.
- Similarly, for conglomerates use the cost of capital for the relevant business, not overall cost of capital.
- Getting the right risk and rate for project can make or break your valuation.
  - Are a **project's business risks** the same as the risks of the other company assets? Is this true **currently**? Is this true **throughout the life of the project**?
  - Will the project support the same capital structure as the company's overall capital structure? Is this true **currently**? Will it remain so **throughout the life of the project**?
- WACC formula assumes that a project will be financed in the same debt-to-equity proportions as the company (or industry) overall.

Valuing Companies with Different Capital Structures

- Companies funded with all equity may want to add debt
- Companies with debt may change their debt policy
- Private company where debt is not observable

Calculate the Unlevered Beta and Levered Beta:

$$\beta_{Unlevered} = \beta_U = \frac{\beta_L}{[1 + (1 - T_C)\frac{D}{E}]}$$

$$\beta_{Levered} = \beta_L = \beta_U[1 + (1 - T_C)\frac{D}{E}]$$

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Costs of Financial Distress

Measuring the Cost of Financial Distress

$$\text{PV (Cost of Financial Distress)} = \frac{Annual\ Probability\ (Financial\ Distress) \times Cost\ of\ Financial\ Distress}{r_d}$$

Measuring the Cost of Financial Distress

$$\text{PV (Cost of Financial Distress)} = \frac{\textit{Annual Probability (Financial Distress)} \times \textit{Cost of Financial Distress}}{r_d}$$

Cost of Financial Distress

- Bankruptcy costs average 7-10% of firm value
- Costs of distress occur well before financial distress and can be larger:
  - Would **customers** still purchase from you (expect lower quality / not honor warranties)?
  - Would **suppliers** still sell to you (at all or only at less favorable terms)?
  - Would high quality **employees** want to stay working for you?
  - Would **cuts** to reduce/avoid financial distress permanently hinder your competitive advantage?

Measuring the Cost of Financial Distress

$$\text{PV (Cost of Financial Distress)} = \frac{\textit{Annual Probability}\ (\textit{Financial Distress}) \times \textit{Cost of Financial Distress}}{r_d}$$

3 Common Ways to Measure Probability of Default

- Corporate Credit Rating (Moody’s / S&P / Fitch)
  - Look at default rates of firms in same category
- Credit Default Swaps (CDS)
  - You generally make a regular payment (think of it as an insurance premium), and the CDS pays out if the firm defaults.
  - $P = 1 - e^{\frac{-S \times t}{1-R}}$, where S is spread (%), t is years to maturity, R is recovery rate (%)

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Measuring the Cost of Financial Distress

3 Common Ways to Measure Probability of Default

- Corporate Credit Rating
- Credit Default Swaps (CDS)
- **Altman Z-Score**
  - Combines 5 business ratios. Each of these is historically higher for average bankrupt (than non-bankrupt) firms two years before bankruptcy.
  - Measure commonly used by auditors, management accountants, courts
  - Z score <1.81 ‘distress zone’ 🚨⚠️, 1.81-2.99 ‘grey zone’ 😅, 3+ ‘safe zone’ 🥳

$$
\begin{aligned}
Z = \; & 0.12 \times (\text{Net Working Capital / Total Assets}) \\
& + 0.14 \times (\text{Retained Earnings / Total Assets}) \\
& + 0.33 \times (\text{Earnings Before Interest and Taxes / Total Assets}) \\
& + 0.006 \times (\text{Market Value of Equity / Book Value of Total Liabilities}) \\
& + 0.999 \times (\text{Sales / Total Assets})
\end{aligned}
$$

Incentive Benefit of Debt

Incentive Benefit of Debt

- **'Agency Problems'** – left to their own devices an agent (e.g., CEO) may maximize their own aims, rather than that of the principal (e.g., shareholders) who the agent works for. Examples:
  - Misuse of firm's funds for private benefits (e.g., excessive executive compensation)
  - Short-term focus / over-risk taking
  - Empire building (make a company large to get more power/prestige even if it doesn't increase shareholder value or creates longer-term issues).

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How does debt help to limit agency problems?

- Debt commits agent to spending money on debt interest (as opposed to discretionary dividends).
- Debt limits how much cash they have available to misuse!

$$V_L = V_U + PV(Tax\ Shield) - PV(Financial\ Distress) - PV\ (Agency)$$

Valuing Firms with Leverage

3 Ways to Value Firms with Leverage

A. Weighted Average Cost of Capital (WACC)

Steps

1. Calculate Free Cash Flows (FCFs)
2. Calculate Weighted Average Cost of Capital (WACC)
3. Discount the FCFs by the WACC

Limitation:

Only considers debt tax shield, not other side effects of debt (distress, agency)

3 Ways to Value Firms with Leverage

B. Adjusted Present Value (APV)

- Value of firm with debt = value of firm without debt + side effects of debt

Steps

1. Value the firm as if all equity
   - Calculate FCF (ignoring leverage) and firm's asset value
2. Value the benefits and costs of debt
   - Calculate the NPV of financing side effects
3. Calculate the value of the leveraged firm

3 Ways to Value Firms with Leverage

C. Flow to Equity Approach (FTE)

- Cashflows from the project to **equity holders** of the levered firm are discounted at the cost of equity capital.

Steps

1. Calculate the levered cash flow (LCF) i.e., cash flow less interest expense
2. Calculate the cost of equity capital
3. Discount the LCF at the cost of equity capital

3 Ways to Value Firms with Leverage. When to Use?

A. Weighted Average Cost of Capital (WACC)
B. Adjusted Present Value (APV)
C. Flow to Equity Approach (FTE)

- Use WACC or FTE if the firm's target debt-to-value ratio applies to the project over its life
- Use APV if the project's level of debt is known over the project's life
- WACC and FTE are more commonly used for capital budgeting,
  but APV may be superior (though hard to measure side-effects of debt)

Financing in the Real-World

Jargon-Busting Long-Term Financing Instruments

1. Common Stock (see Week 3)

2. Preferred Stock (see Week 3)

3. Long-Term Debt (see Week 2)

- Long-term debt is 1+ year long (otherwise ‘short-term debt’)
- No voting rights
- Can be fixed or floating interest rates
- Can be secured (collateral are assets) / mortgage (collateral is property) / debentures (no collateral, 10+ years, may convert to shares)
- Can have other covenants
- Most bonds are ‘callable’ (bond issuer has right to buy back the bond before maturity at pre-determined price)
- Syndicated loans (many lenders jointly finance a single borrower. This reduces has risk sharing benefits from reducing each lender’s exposure.)
- Leveraged loans (loans to finance acquisitions or buyouts)

Where to Find External Long-Term Financing

- **Commercial Banks for Debt Financing**

  - banks have a hands-on role screening the firm's creditworthiness and monitoring its ability to repay debt.

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  - Issuing public debt is similar to equity, but less administration costs to issue, and 'private placements' more common than equity to sell directly to a small group of institutional investors e.g., pension funds)

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- **Equity Capital Markets**
  - IPOs. Seasoned Equity Offerings (SEOs) where already public company issues more shares (also known as secondary offering).

Do CFOs Think That Their Stock is Correctly Valued?

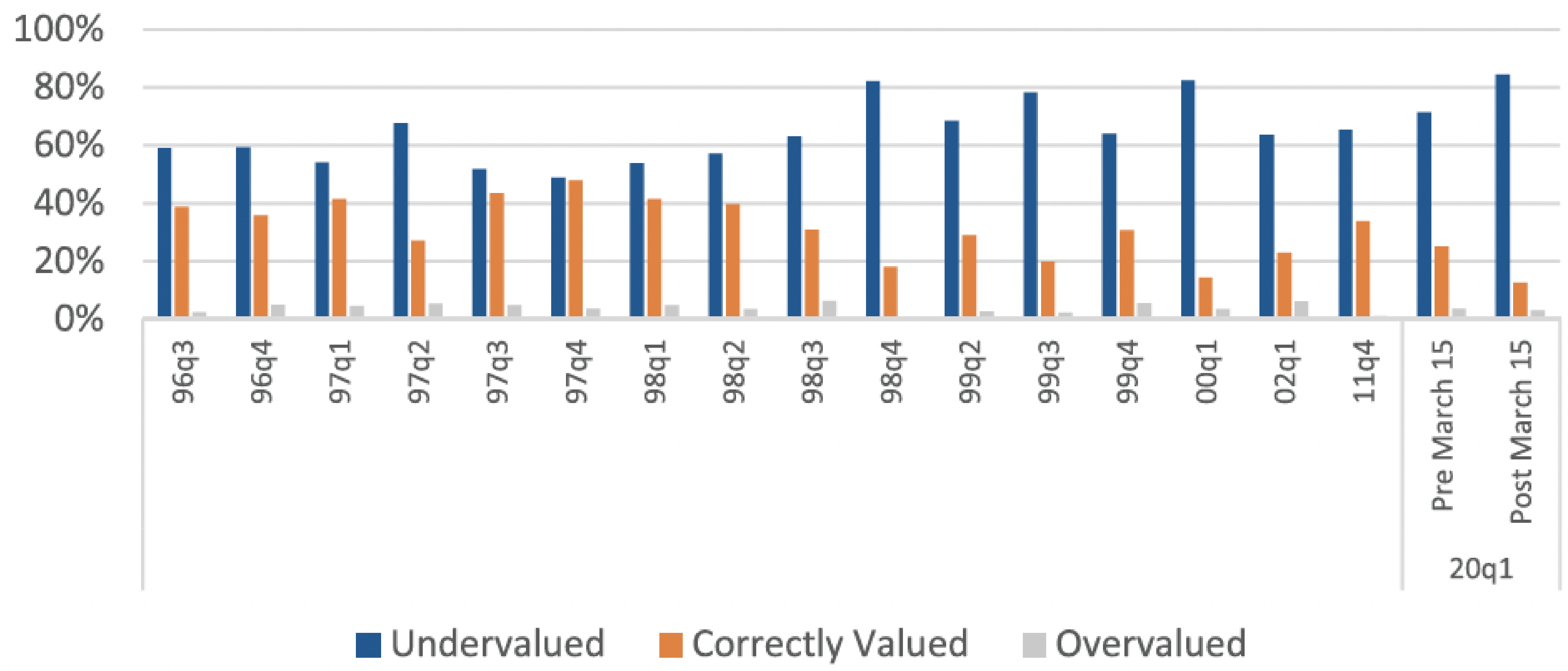

Figure 18. Is your stock correctly valued? (1990s, 2000, 2002, 2011, 2020). This figure displays CFO responses to the following question: *Is your stock correctly valued?* Blue (left) bars display the percentage of CFOs who believe that their firm's common stock is undervalued. Orange (middle) bars display results for CFOs who believe that their stock is correctly valued. Gray (right) bars display results for CFOs who believe that their stock is overvalued. The historic data are from the Duke Global Business Outlook survey. 20Q1 data are from the March 2020 wave of the 2022 survey. For example, among CFOs who responded after March 15, 2020, 83% think that their stocks are undervalued. (Color figure can be viewed at wileyonlinelibrary.com)

Related to the **Pecking Order Theory (Myers & Majluf, 1984)** of capital structure that managers prefer to finance investments with the company's retained earnings, and if not, then with debt, and only issue equity as a last resort.

How Do CFOs Measure Leverage?

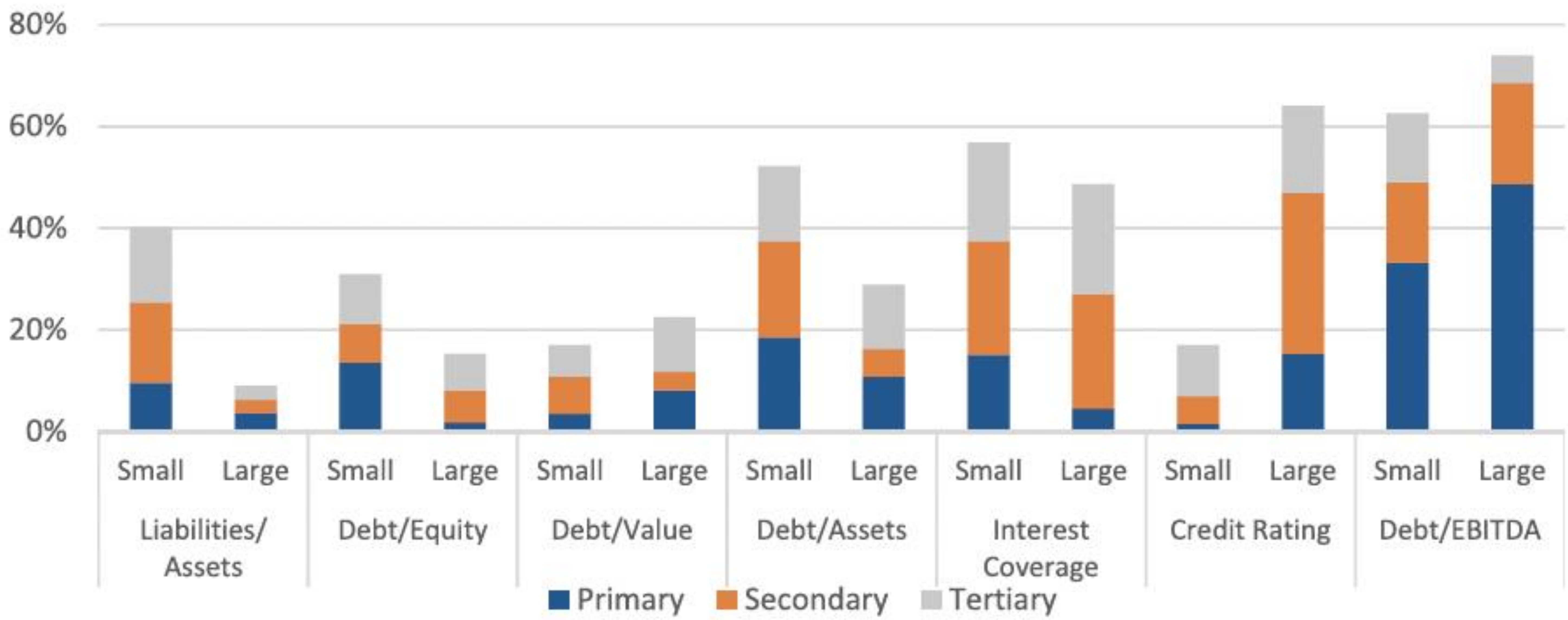

Figure 13. How do companies measure leverage? This figure displays CFO responses to the following question: *When you consider the appropriate amount of debt for your firm (optimal capital structure), what are the primary metrics your company uses? (rank your top 3)* The blue bottom bars display results for the primary choice. The orange middle (gray top) bars display results for the secondary (tertiary) choices, respectively. The results are presented conditional on firm size. Large firms have annual revenue greater than $1 billion, and small firms have annual revenue less than $1 billion. For example, 49% of CFOs from large firms indicate that their primary measure of capital structure is debt/EBITDA (and 74% say that it is one of their top three debt measures). (Color figure can be viewed at wileyonlinelibrary.com)

CFO Target Debt Ratio and Time to Return to Target Debt Ratio

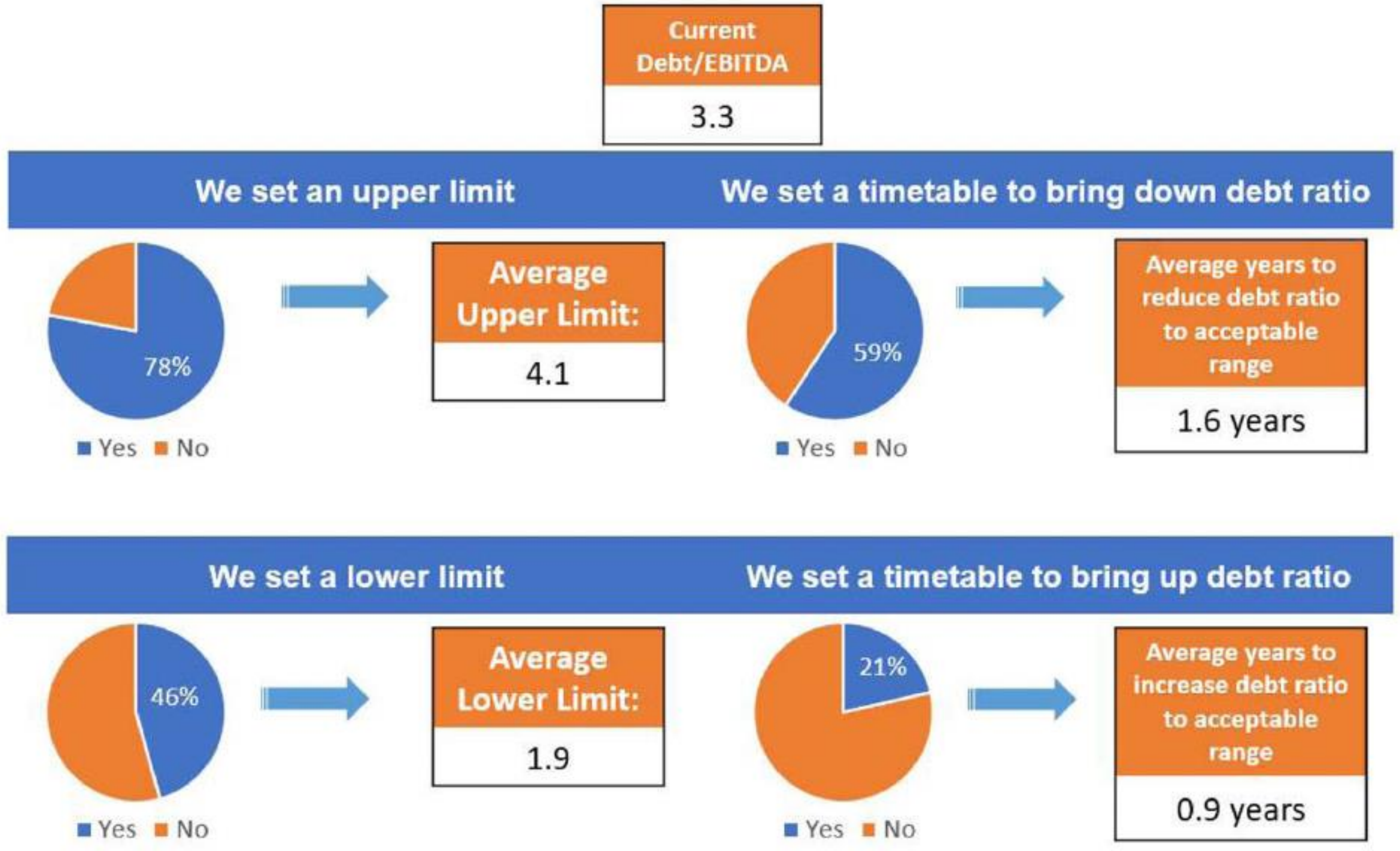

Figure 15. Debt ratio ranges and timetable to return to target. This figure displays March 2019 CFO responses to questions about whether they set an upper or lower bound as part of an acceptable range for their target debt ratios. These responses are only displayed for firms that indicated that they had a strict, somewhat tight, or flexible debt range (in Figure 14), among firms that indicated debt/EBITDA was their primary debt metric (Figure 13). Mean debt/EBITDA was 3.3 among these firms at the time of the survey. Of these firms, 78% indicated that they set an upper-limit debt ratio, with a mean upper limit of 4.1, and 59% of these firms set a timetable to reduce their debt ratio when it hit the upper limit, with a mean of 1.6 years. The lower limit results shown at the bottom are interpreted analogously. (Color figure can be viewed at wileyonlinelibrary.com)

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Which Factors Drive CFO Decisions on How Much Debt to Use?

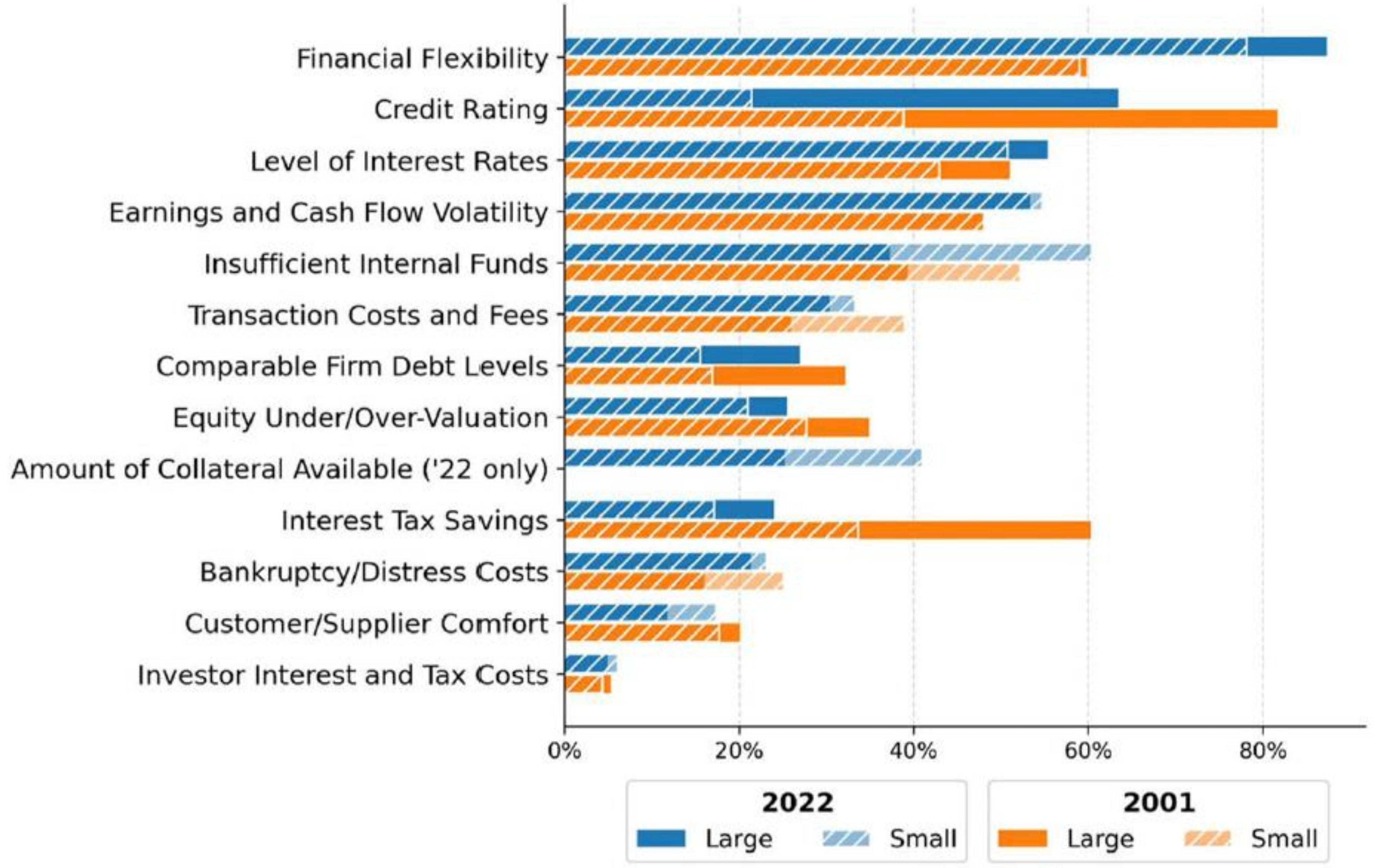

Figure 16. Which factors drive debt decisions? This figure displays CFO responses to the following question: *Which of the following factors affect how your firm chooses the appropriate amount of debt for your firm? {0 = Not Important, 1, 2 = Moderate Importance, 3, 4 = Very Important.* The percentage of firms that answer "3" or "4" is shown. Blue (top) bars display results for the 2022 Duke CFO survey (March 2019 wave); orange (bottom) bars display results from Graham and Harvey (2001). Within each blue and orange bar, the solid portion displays responses for large firms (revenue above $1 billion), and the crosshatched portion displays responses for small firms (revenue below $1 billion). For example, 87% of large firms (78% of small firms) in 2022 regard maintaining financial flexibility as an important or very important factor affecting debt decisions. The 2022 credit rating number in the figure (i.e., 63.5% for large firms) is for firms that indicated that they had a credit rating on the survey. For the firms that I can confirm have a Standard & Poor's credit rating, the percentage that listed credit rating as important or very important is 71.9% for the full sample. (Color figure can be viewed at wileyonlinelibrary.com)

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Why Do CFOs Think That Maintaining Financial Flexibility Is Important?

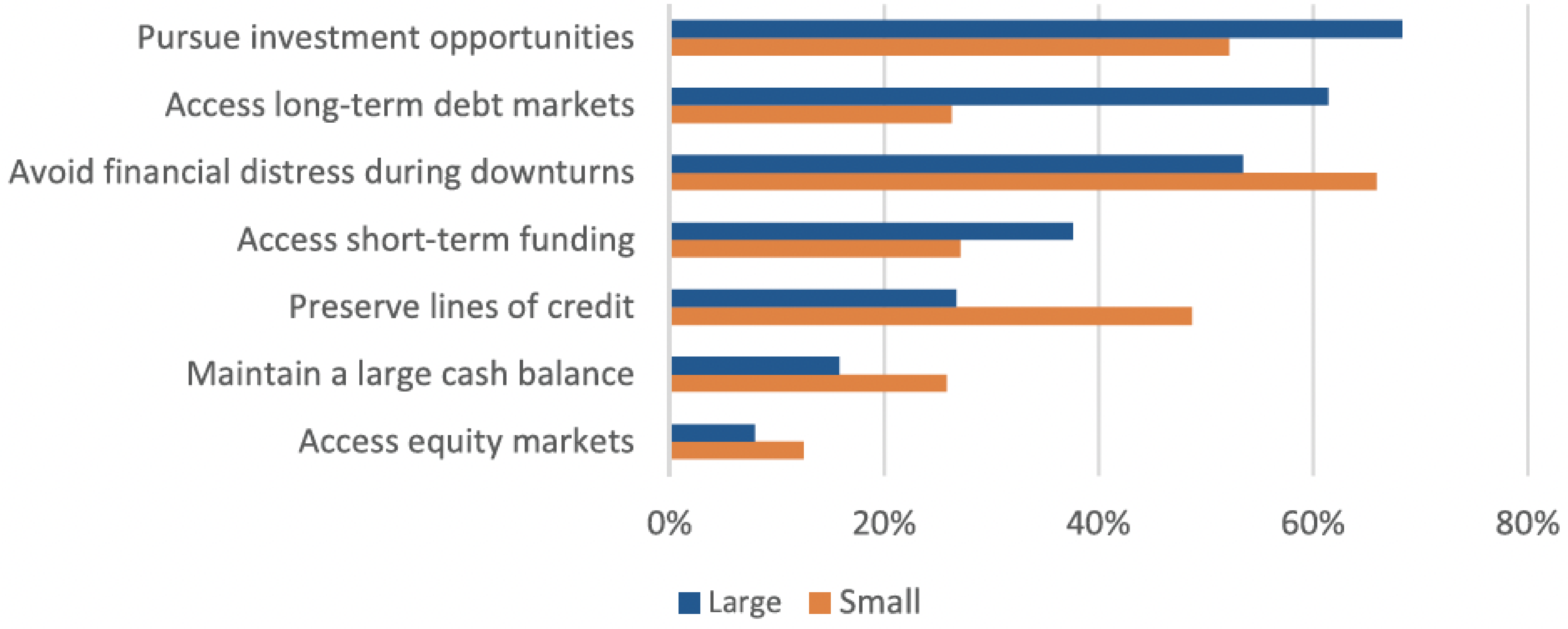

Figure 17. Why is maintaining financial flexibility important? This figure displays the 2022 CFO responses to the following question: *Why is it important for your firm to maintain financial flexibility? (choose up to three)* This question was only asked of firms that indicated that financial flexibility was at least moderately important (answered "2," "3," or "4") in Figure 16). Large (small) firms are those with sales revenue greater (less) than $1 billion. Blue (top) bars display responses from large firms; orange (bottom) bars display responses for small firms. For example, two-thirds of small firms indicate that financial flexibility is important to help avoid financial distress. (Color figure can be viewed at wileyonlinelibrary.com)

Examples

Q1. True or False under MM Theory without taxes? “As the firm borrows more and debt becomes risky, both stock- and bondholders demand higher rates of return. Thus, by reducing the debt ratio we can reduce both the cost of debt and the cost of equity, making everybody better off.”

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Q1. True or False under MM Theory without taxes? "As the firm borrows more and debt becomes risky, both stockholders and bondholders demand higher rates of return. Thus, by reducing the debt ratio we can reduce both the cost of debt and the cost of equity, making everybody better off."

FALSE. The firm's cost of capital is invariant to the capital structure if taxes=0.

Q2. True or False under MM Theory without taxes?
“Moderate borrowing doesn't significantly affect the probability of financial distress or bankruptcy. Consequently, moderate borrowing won't increase the expected rate of return demanded by stockholders.”

Q2. True or False under MM Theory without taxes? "Moderate borrowing doesn't significantly affect the probability of financial distress or bankruptcy. Consequently, moderate borrowing won't increase the expected rate of return demanded by stockholders."

FALSE. Leverage always increases the risk to stockholders, who in turn will ask for a higher rate of return.

Q3. GK Airlines is currently an unlevered firm. It is considering a capital restructuring to allow $200 of debt. The company expects to generate $151.52 in cash flows before interest and taxes, in perpetuity. The corporate tax rate is 34 percent. Its cost of debt capital is 10 percent. Unlevered firms in the same industry have a cost of equity capital of 20 percent. What is the value of GK Airlines without debt. What will the new value of GK Airlines be with debt?

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EBIT = 151.52 in perpetuity

$200 of debt. Tax = 34%. r_D=10%. $r_A = r_U$ = 20%

Old Value: $151.52 x (1 – 0.34) / 0.2 = $500. We are taking the pre-tax cash flow, adjusting for taxes, and discounting it.

Value of Debt Tax Shield = $200 x 0.34 = $68. What about r_D? We could write out the present value of the tax shield as: ($200 x 0.34 x 0.1) / 0.1 so r_D is on the numerator and denominator and cancels out.

New firm value = $V_U + T_C D$ = $500 + $68 = $568

Suppose that the question also asked for the cost of equity after adding debt. How would we calculate that?

$$r_E = r_U + \frac{D}{E}(r_U - r_D)(1 - T_c) = 0.2 + \left(\frac{200}{568 - 200}\right) \times (0.2 - 0.1) \times (1 - 0.34) = 23.59\%$$

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Q4. Calculate the overall cost of capital given the below balance sheet (market values). Assume the marginal tax rate is 30%. r_D is 7.5% and r_E is 15%.

Asset Value: \$100	Debt (D): \$30
	Equity (E): \$70
Asset Value: \$100	**Firm Value: \$100**

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Q4. Calculate the overall cost of capital given the below balance sheet (market values). Assume the marginal tax rate is 30%. r_D is 7.5% and r_E is 15%.

Asset Value: $100	Debt (D): $30
	Equity (E): $70
Asset Value: $100	**Firm Value: $100**

WACC = 7.5% x (1 – 30%) x ($30/$100) + 15% x ($70/$100) = 12.075%

Q5. Suppose is funded by common shares, preferred shares, an debt. The company is funded by 50% common shares and 10% preferred shares. The beta for the common shares, preferred shares, and debt are 1.4, 0.9, and 0.2. Assume the risk-free rate is 3% and the market return is 11%. What is the WACC of the firm?

Q5. Suppose is funded by common shares, preferred shares, and debt. The company is funded by 50% common shares and 10% preferred shares. The beta for the common shares, preferred shares, and debt are 1.4, 0.9, and 0.2. Assume the risk-free rate is 3% and the market return is 11%. What is the WACC of the firm?

$$E[r_D] = r_{rf} + \beta_D\left(E[r_M] - r_{rf}\right) = 4.6\%$$

$$E[r_P] = r_{rf} + \beta_P\left(E[r_M] - r_{rf}\right) = 10.2\%$$

$$E[r_E] = r_{rf} + \beta_E\left(E[r_M] - r_{rf}\right) = 14.2\%$$

$$WACC = (1 - T_c)\, r_D \left(\frac{D}{V}\right) + r_P\left(\frac{P}{V}\right) + r_E\left(\frac{E}{V}\right)$$

WACC = (1 - 0.30) x 0.046 x 0.40 + 0.102 x 0.10 + 0.142 x 0.50 = 9.408%

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ness

Q6. Trombone Corporation has a current stock price of $7.50 per share, and has 100 million outstanding shares. The cost of equity is 12.4%. The stock is trading above its book value of $5.00 per share. Interest rates have been stable since the firm's $500 million debt was issued and so the firm's book and market values of debt are the same, and costs 6%. The firm's assets are not traded and so not observed. The tax rate is 35%. What is the WACC of the firm?

Q6. Trombone Corporation has a current stock price of \$7.50 per share, and has 100 million outstanding shares. The cost of equity is 12.4%. The stock is trading above its book value of \$5.00 per share. Interest rates have been stable since the firm's \$500 million debt was issued and so the firm's book and market values of debt are the same, and costs 6%. The firm's assets are not traded and so not observed. The tax rate is 35%. What is the WACC of the firm?

E = \$7.5 x 100 = \$750

V = \$500 + \$750 = \$1,250

D/V = \$500/\$1,250 = 0.4

E/V = \$750/\$1,250 = 0.6

WACC = (1 - 0.35) x 0.06 x 0.4 + 0.124 x 0.6 = 0.090 or 9.0%

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Q7. Using your answer from the previous question, Trombone Corporation is now planning a $12.5 million investment in a trombone making machine. It never depreciates. It generates a perpetual stream of $1.731 million per year pre-tax cashflows. What’s the value of the project?

Q7. Using your answer from the previous question, Trombone Corporation is now planning a $12.5 million investment in a trombone making machine. It never depreciates. It generates a perpetual stream of $1.731 million per year pre-tax cashflows. What's the value of the project?

WACC = 9.0%

After-tax cashflow = $1.731 x 0.65 = $1.12515

NPV = -$12.5 + 1.12515/0.09 = 0.0017

What about the tax shield? That's being reflected in the WACC

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Q8. Cool Conglomerate just hired you as its financial manager. The conglomerate consists of 3 divisions of equal size: Automotive, Music, and Utilities. The beta of automotive is 2.0, the beta of music is 1.3, and the beta of utilities is 0.6. Assume that the risk-free rate is 4% and the excess return is 10%.

A) What is the company's beta?

B) What is the company's expected return?

C) What is the cost of capital to evaluate a new automative investment?

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A) What is the company's beta?

B) What is the company's expected return?

C) What is the cost of capital to evaluate a new automative investment?

A) β = 1/3 x 2 + 1/3 x 1.3 + 1/3 x 0.6 = 1.3

B) E[r] = $r_{rf} + \beta \ (r_m - r_{rf}) = 4\% + 1.3$ x 10% = 17%

C) Use rate for automotives not entire firm!

E[r_{auto}] = $r_{rf} + \beta_{auto}(r_m - r_{rf}) = 4\% + 2$ x 10% = 24%

Q9. Physical Education a private equity company trying to determine its cost of equity. The average beta is 0.9 in a list of comparable firms (in its education industry).Comparable firms have an average debt-to-equity ratio of 0.5. Physical Education has a debt-to-equity ratio of 25%. The tax rate is 30%, the expected market return is 10%, the risk free rate is 3%, and the cost of debt is 5%.
What is Physical Education's cost of equity?

Q9. Physical Education a private equity company trying to determine its cost of equity. The average beta is 0.9 in a list of comparable firms (in its education industry).Comparable firms have an average debt-to-equity ratio of 0.5. Physical Education has a debt-to-equity ratio of 25%. The tax rate is 30%, the expected market return is 10%, the risk free rate is 3%, and the cost of debt is 5%.
What is Physical Education's cost of equity?

$$\beta_{Unlevered} = \beta_U = \frac{\beta_L}{[1 + (1 - T_C)\frac{D}{E}]} = \frac{0.9}{[1 + (1 - 0.3)\ \times\ 0.5]} = 0.67$$

$$\beta_{Levered} = \beta_L = \beta_U \left[1 + (1 - T_C)\frac{D}{E}\right] = 0.67[1 + (1 - 0.3)\ \times\ 0.25] = 0.79$$

$$E[r_i] = r_{rf} + \beta_L\left(E[r_m] - r_{rf}\right) = 0.03 + 0.79\ \times\ (0.10 - 0.03) = 8.53\%$$